INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

# MICROCIRCUITS, DIGITAL, BIPOLAR, PROGRAMMABLE LOGIC, MONOLITHIC SILICON

Inactive for new design after 28 July 1995

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, bipolar, programmable logic microcircuits which employ titanium tungsten (Ti-W) as the fusible link or programming element. A choice of case outlines and lead materials and finishes are provided for each type and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.4).

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535.

1.2.1 Device types. The device types are as follows:

| Device type | Circuit |
|---|---|
| 01 | 16-input 8-output and/or invert gate array. |
| 02 | 16-input 8-output registered and/or gate array. |
| 03 | 16-input 6-output registered and/or gate array. |
| 04 | 16-input 4-output registered and/or gate array. |
| 05 | 16-input 8-output and/or invert gate array. |
| 06 | 16-input 8-output registered and/or gate array. |
| 07 | 16-input 6-output registered and/or gate array. |
| 08 | 16-input 4-output registered and/or gate array. |
| 09 | 16-input 8-output and/or invert gate array. |
| 10 | 16-input 8-output registered and/or gate array. |
| 11 | 16-input 6-output registered and/or gate array. |
| 12 | 16-input 4-output registered and/or gate array. |

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

| Outline letter | Descriptive designator | Terminals | Package style |
|---|---|---|---|
| R | GDIP1-T20 or CDIP2-T20 | 20 | Dual-in-line |
| 2 | CQCC1-N20 | 20 | Square chip carrier |

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P. O. Box 3990, Columbus, OH 43218-3990, or emailed to memory@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil



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1.3 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range | -0.5 V dc to +10.75 V |
| Input voltage range | -1.5 V dc to +10.75 V |
| Storage temperature range | -65° to +150°C |
| Lead temperature (soldering, 10 seconds) | +260°C |
| Thermal resistance, junction to case ($\theta_{JC}$): 1/ 2/ | See MIL-STD-1835 |
| Output voltage applied | -1.5 V to +10.75 V |
| Maximum power dissipation ($P_D$): | |
| Device types 01, 02, 03, 04 | 1.1 W |
| Device types 05, 06, 07, 08 | 0.6 W |
| Device types 09, 10, 11, 12 | 1.2 W |
| Maximum junction temperature ($T_J$) | 175°C |

1.4 Recommended operating conditions.

| | |
|---|---|
| Supply voltage ($V_{CC}$) | 4.5 V minimum to 5.5 V maximum |
| Minimum high-level input voltage ($V_{IH}$) | 2.0 V dc |
| Maximum low-level input voltage ($V_{IL}$) | 0.8 V dc |
| Case operating temperature range ($T_C$) | -55 °C to +125 °C |

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

1/ Heat sinking is recommended to reduce the junction temperature.
2/ Must withstand the added $P_D$ due to short circuit test (e.g. $I_{OS}$).



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related specification sheets), the text of this specification takes precedence. Nothing in this specification, however supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.3).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections and logic diagrams shall be as specified on figure 1.

3.3.2 Truth table.

3.3.2.1 Unprogrammed devices. The truth table for unprogrammed devices for contracts involving no altered item drawing shall be as specified on figure 2. When required in groups A, B, or C (see 4.4), the devices shall be programmed by the manufacturer prior to test. A minimum of 50 percent of the total number of fuses shall be programmed or to any altered item drawing pattern which programs at least 25 percent of the total number of fuses.

3.3.2.2 Programmed devices The truth table for programmed devices shall be as specified by the altered item drawing.

3.3.3 Logic diagram. The logic diagram for unprogrammed devices shall be as specified on figure 3.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table II and (where applicable), the altered item drawing for the applicable device type and device class. The subgroups of table III which constitue the minimum electrical test requirements for screening, qualification, and quality conformance, by device class, are specified in table II.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535. For programmed devices, the altered item drawing number shall be added to the marking by the programming activity.

3.8 Processing options. Since the device is capable of being programmed by either the manufacturer or the user to result in a wide variety of configurations, two processing options are provided for selection in the contract, using an altered item drawing.

3.8.1 Unprogrammed device delivered to the user. All testing shall be verified through group A testing as defined in 3.3.2.1, tables II and III. It is recommended that users perform subgroups 7 and 9 after programming to verify the specific program configuration.

3.8.2 Manufacturer-programmed device delivered to the user. All testing requirements and quality assurance provisions herein, including the requirements of the altered item drawing, shall be satisfied by the manufacturer prior to delivery.

3.9 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 14 (see MIL-PRF-38535.)



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TABLE I. Electrical performance characteristics.

| Test | Symbol | Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified | | Device Type | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V, $I_I$ = -18 mA | | All | | -1.5 | V |
| High-level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V $V_{IL}$ = 0.0 V<br>$V_{IH}$ = 3.0 V, $I_{OH}$ = -2 mA | | All | 2.4 | | V |
| Low-level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V $V_{IL}$ = 0.0 V<br>$V_{IH}$ = 3.0 V, $I_{OL}$ = 12 mA | | All | | 0.5 | V |
| High level input current | $I_{IH}$ | $V_{CC}$ = 5.5 V 2/<br>$V_I$ = 2.7 V | Pin 1, 11 | All | | 50 | µA |
| | | | All others | All | | 20 | µA |
| Low level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V $V_I$ = 0.4 V 2/ | | All | | -.25 | mA |
| Output short circuit current | $I_O$ | $V_{CC}$ = 5.5 V $V_O$ = 2.25 V 3/ | | All | -15 | -125 | mA |
| Input current | $I_I$ | $V_{CC}$ = 5.5 V, $V_I$ = 5.5 V | $V_{OH}$ = 1, 11 | All | | 0.2 | mA |
| | | | All others | All | | 0.1 | mA |
| Off-state output current | $I_{OZL}$ | $V_{CC}$ = 5.5 V, $V_{IL}$ = 0.0 V<br>$V_{IH}$ = 3.0 V, $V_O$ = 0.4 V 2/ | Outputs | All | | -20 | µA |
| | | | I/O ports | All | | -250 | µA |
| Off-state ouput current | $I_{OZH}$ | $V_{CC}$ = 5.5 V, $V_{IL}$ = 0.0 V<br>$V_{IH}$ = 3.0 V, $V_O$ = 2.7 V | Outputs | All | | 20 | µA |
| | | | I/O ports | All | | 100 | µA |
| Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V<br>$V_I$ = 0.0 V | Outputs open | 01, 02, 03, 04 | | 190 | mA |
| | | | | 05, 06, 07, 08 | | 105 | |
| | | | | 09, 10, 11, 12 | | 220 | |
| Propagation delay I, I/O, to 0, I/O | $t_{PHL}$ | See figure 4 4/ | | 01, 03, 04 | | 20 | ns |
| | | | | 05, 07, 08 | | 30 | |
| | | | | 09, 11, 12 | | 15 | |
| Propagation delay I, I/O, to 0, I/O | $t_{PLH}$ | | | 01, 03, 04 | | 20 | ns |
| | | | | 05, 07, 08 | | 30 | |
| | | | | 09, 11, 12 | | 15 | |
| Enable time I, I/O, to 0, I/O | $t_{PZH}$ | | | 01, 03, 04 | | 20 | ns |
| | | | | 05, 07, 08 | | 30 | |
| | | | | 09, 11, 12 | | 15 | |
| Enable time I, I/O, to 0, I/O | $t_{PZL}$ | | | 01, 03, 04 | | 20 | ns |
| | | | | 05, 07, 08 | | 30 | |
| | | | | 09, 11, 12 | | 15 | |
| Disable time I, I/O, to 0, I/O | $t_{PHZ}$ | | | 01, 03, 04 | | 20 | ns |
| | | | | 05, 07, 08 | | 30 | |
| | | | | 09, 11, 12 | | 15 | |
| Disable time I, I/O, to 0, I/O | $t_{PLZ}$ | | | 01, 03, 04 | | 20 | ns |
| | | | | 05, 07, 08 | | 30 | |
| | | | | 09, 11, 12 | | 15 | |
| Enable time /OE to Q | $t_{PZH}$ | See figure 4 5/ | | 02, 03, 04 | | 15 | ns |
| | | | | 06, 07, 08 | | 25 | |
| | | | | 10, 11, 12 | | 12 | |
| Enable time /OE to Q | $t_{PZL}$ | | | 02, 03, 04 | | 15 | ns |
| | | | | 06, 07, 08 | | 25 | |
| | | | | 10, 11, 12 | | 12 | |



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TABLE I. Electrical performance characteristics.

| Test | Symbol | Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified | Device Type | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Disable time /OE to Q | $t_{PHZ}$ | See figure 4 5/ | 02, 03, 04 | | 15 | ns |
| | | | 06, 07, 08 | | 25 | |
| | | | 10, 11, 12 | | 12 | |
| Disable time /OE to Q | $t_{PLZ}$ | | 02, 03, 04 | | 15 | |
| | | | 06, 07, 08 | | 25 | |
| | | | 10, 11, 12 | | 12 | |
| Propagation delay clock to Q | $t_{PCH}$ | | 02, 03, 04 | | 15 | ns |
| | | | 06, 07, 08 | | 20 | |
| | | | 10, 11, 12 | | 12 | |
| Propagation delay clock to Q | $t_{PCL}$ | | 02, 03, 04 | | 15 | ns |
| | | | 06, 07, 08 | | 20 | |
| | | | 10, 11, 12 | | 12 | |
| Minimum clock pulse | $t_{p(CL)}$ | | 02,03, 04 | 12 | | ns |
| | | | 06, 07, 08 | 20 | | |
| | | | 10, 11, 12 | 10 | | |
| Minimum setup time | $t_{SU}$ | | 02, 03, 04 | 20 | | ns |
| | | | 06, 07, 08 | 25 | | |
| | | | 10, 11, 12 | 15 | | |
| Hold time | $t_H$ | | All | 0 | | ns |
| Maximum clock frequency I, I/O, to 0, I/O | $f_{MAX}$ | | 02, 03, 04 | 40 | | MHz |
| | | | 06, 07, 08 | 25 | | |
| | | | 10, 11, 12 | 50 | | |

1/ Complete terminal conditions shall be specified in table III.
2/ I/O terminal leakage is the worst case $I_{IX}$ or $I_{OZH}$
3/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current , $I_O$. Only one output tested at a time.
4/ Applies to non-registered output only, with internal output enables.
5/ Applies to registered outputs only, with external output enable (pin 11).



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TABLE II. Electrical test requirements. 1/ 2/ 3/

| MIL-PRF-38535 test requirements | Subgroups (see table III) | |
|---|---|---|
| | Class B devices | Class S devices |
| Interim electrical parameters | 1 | 1 |
| Final electrical test parameters for unprogrammed devices | 1*, 2, 3, 7*, 8 | 1*, 2, 3, 7*, 8 |
| Final electrical test parameters for programmed devices | 1*, 2, 3, 7*, 8, 9 | 1*, 2, 3, 7*, 8, 9, 10, 11 |
| Group A test requirements | 1, 2, 3, 7, 8, 9, 10, 11 | 1, 2, 3, 7, 8 9, 10, 11 |
| Group B end-point electrical parameters subgroup 5 | N/A | 1, 2, 3, 7, 8, 9, 10, 11 |
| Group C end-point electrical parameters | 1, 2, 3, 7, 8 | 1, 2, 3, 7, 8, 9, 10, 11 |
| Group D end-point electrical parameters | 1, 2, 3, 7, 8 | 1, 2, 3, 7, 8 |

1/ (*) indicates PDA applies to subgroups 1 and 7.
2/ Any or all subgroups may be combined when using high-speed testers.
3/ Subgroup 7 and 8 shall consist of verifying the pattern specified.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535, appendix B.

d. All devices processed by the manufacturer to an altered item drawing shall be programmed prior to burn-in.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).



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4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

c. For unprogrammed devices, a sample shall be selected to satisfy programmability requirements prior to performing subgroups 9, 10, and 11. Twelve devices shall be submitted to programming (see 3.3.2.1). If more than two devices fail to program, the lot shall be rejected. At the manufacturers option, the sample may be increased to 24 total devices with no more than four total device failures allowable.

d. For unprogrammed devices, 10 devices from the programmability sample shall be submitted to the requirements of the specified tests of subgroups 1, 2, and 3 designated for programmed devices only. If any device fails, the lot shall be rejected. At the manufacturer s option, the sample may be increased to 20 total devices with no more than two total device failures allowable.

e. For unprogrammed devices, 10 devices from the programmability sample shall be submitted to the requirements of group A, subgroups 9, 10, and 11. If more than two total devces fail the lot shall be rejected. At the manufacturers option, the sample may be increased to 20 total devices with no more than four total failures allowable.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II MIL-PRF-38535.

a. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

c. For qualification, at least 50 percent of the sample selected for life testing shall be programmed (see 3.3.2). For quality conformance inspection, the programmability sample (see 4.4.1c) shall be included in the life test.



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| Pin number | Device types 01, 05, and 09 | Device types 01, 05, and 09 |
|---|---|---|
| | CASES | |
| | R | 2 |
| 1 | I | I |
| 2 | I | I |
| 3 | I | I |
| 4 | I | I |
| 5 | I | I |
| 6 | I | I |
| 7 | I | I |
| 8 | I | I |
| 9 | I | I |
| 10 | GND | GND |
| 11 | I | I |
| 12 | O | O |
| 13 | I/O | I/O |
| 14 | I/O | I/O |
| 15 | I/O | I/O |
| 16 | I/O | I/O |
| 17 | I/O | I/O |
| 18 | I/O | I/O |
| 19 | O | O |
| 20 | $V_{CC}$ | $V_{CC}$ |

FIGURE 2. Terminal connections.



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| Pin number | Device types 02, 06, and 10 | Device types 02, 06, and 10 |
|---|---|---|
| | CASES | |
| | R | 2 |
| 1 | CLK | CLK |
| 2 | I | I |
| 3 | I | I |
| 4 | I | I |
| 5 | I | I |
| 6 | I | I |
| 7 | I | I |
| 8 | I | I |
| 9 | I | I |
| 10 | GND | GND |
| 11 | $\overline{OE}$ | $\overline{OE}$ |
| 12 | O | O |
| 13 | O | O |
| 14 | O | O |
| 15 | O | O |
| 16 | O | O |
| 17 | O | O |
| 18 | O | O |
| 19 | O | O |
| 20 | $V_{CC}$ | $V_{CC}$ |

FIGURE 2. Terminal connections – Continued.



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| Pin number | Device types 03, 07, and 11 | Device types 03, 07, and 11 |
|---|---|---|
| | CASES | |
| | R | 2 |
| 1 | CLK | CLK |
| 2 | I | I |
| 3 | I | I |
| 4 | I | I |
| 5 | I | I |
| 6 | I | I |
| 7 | I | I |
| 8 | I | I |
| 9 | I | I |
| 10 | GND | GND |
| 11 | $\overline{OE}$ | $\overline{OE}$ |
| 12 | I/O | I/O |
| 13 | O | O |
| 14 | O | O |
| 15 | O | O |
| 16 | O | O |
| 17 | O | O |
| 18 | O | O |
| 19 | I/O | I/O |
| 20 | $V_{CC}$ | $V_{CC}$ |

FIGURE 2. Terminal connections – Continued.



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| Pin number | Device types 04, 08, and 12 | Device types 04, 08, and 12 |
|---|---|---|
| | CASES | |
| | R | 2 |
| 1 | CLK | CLK |
| 2 | I | I |
| 3 | I | I |
| 4 | I | I |
| 5 | I | I |
| 6 | I | I |
| 7 | I | I |
| 8 | I | I |
| 9 | I | I |
| 10 | GND | GND |
| 11 | $\overline{OE}$ | $\overline{OE}$ |
| 12 | I/O | I/O |
| 13 | I/O | I/O |
| 14 | O | O |
| 15 | O | O |
| 16 | O | O |
| 17 | O | O |
| 18 | I/O | I/O |
| 19 | I/O | I/O |
| 20 | $V_{CC}$ | $V_{CC}$ |

FIGURE 2. Terminal connections – Continued.



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Device types 01, 05, and 09

| Truth table | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Inputs | | | | | | | | | | Outputs | | | | | | | |
| $I_9$ | $I_8$ | $I_7$ | $I_6$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | $D_7$ | $I/D_6$ | $I/D_5$ | $I/D_4$ | $I/D_3$ | $I/D_2$ | $I/D_1$ | $D_0$ |
| X | X | X | X | X | X | X | X | X | X | Z | Z | Z | Z | Z | Z | Z | Z |

Device types 02, 06, and 10

| Truth table | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Inputs | | | | | | | | | | Outputs | | | | | | | |
| OE | $I_7$ | $I_6$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | CLK | $D_7$ | $D_6$ | $D_5$ | $D_4$ | $D_3$ | $D_2$ | $D_1$ | $D_0$ |
| H | X | X | X | X | X | X | X | X | CLK | Z | Z | Z | Z | Z | Z | Z | Z |
| L | X | X | X | X | X | X | X | X | CLK | H | H | H | H | H | H | H | H |

Device types 03, 07, and 11

| Truth table | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Inputs | | | | | | | | | | Outputs | | | | | | | |
| OE | $I_7$ | $I_6$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | CLK | $I/D_7$ | $D_6$ | $D_5$ | $D_4$ | $D_3$ | $D_2$ | $D_1$ | $I/D_0$ |
| H | X | X | X | X | X | X | X | X | CLK | Z | Z | Z | Z | Z | Z | Z | Z |
| L | X | X | X | X | X | X | X | X | CLK | Z | H | H | H | H | H | H | Z |

Device types 04, 08, and 12

| Truth table | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Inputs | | | | | | | | | | Outputs | | | | | | | |
| OE | $I_7$ | $I_6$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | CLK | $I/D_7$ | $I/D_6$ | $D_5$ | $D_4$ | $D_3$ | $D_2$ | $I/D_1$ | $I/D_0$ |
| H | X | X | X | X | X | X | X | X | CLK | Z | Z | Z | Z | Z | Z | Z | Z |
| L | X | X | X | X | X | X | X | X | CLK | Z | Z | H | H | H | H | Z | Z |

FIGURE 2. Truth table (unprogrammed).



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DEVICE TYPES 01,05 AND 09

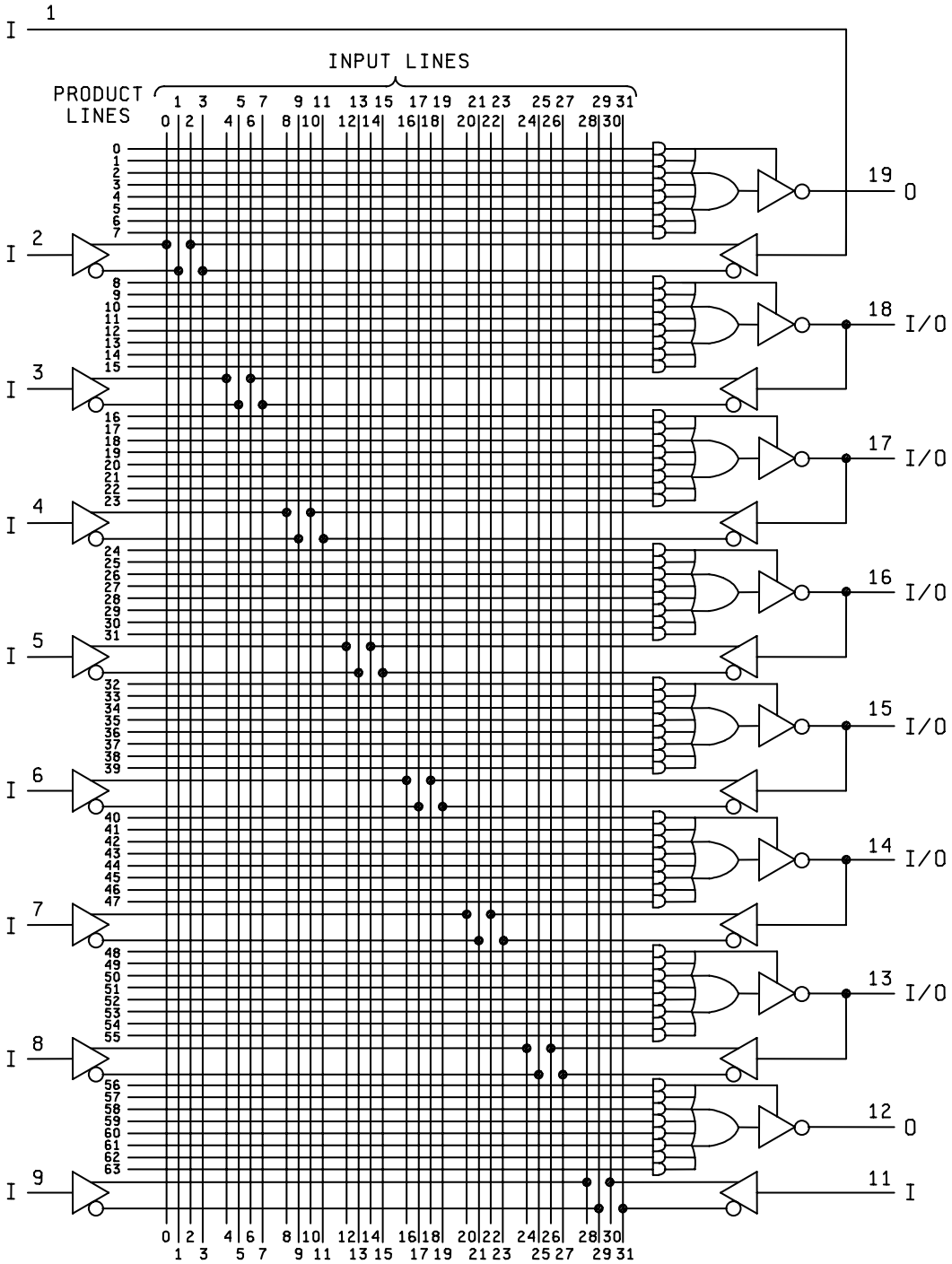



FIGURE 3. Logic diagrams (unprogrammed).



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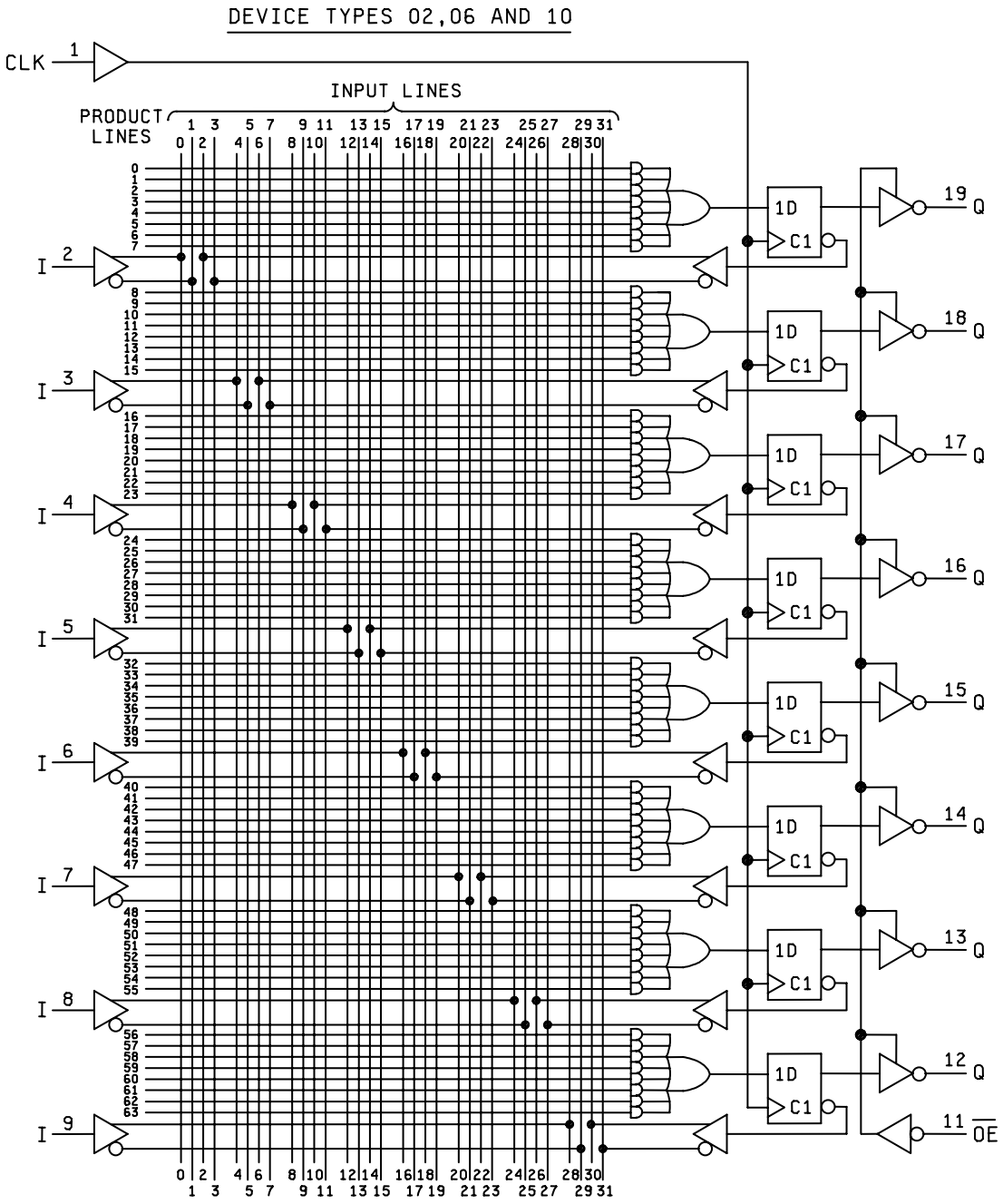



FIGURE 3. Logic diagrams (unprogrammed) – Continued.



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DEVICE TYPE 03,07 AND 11

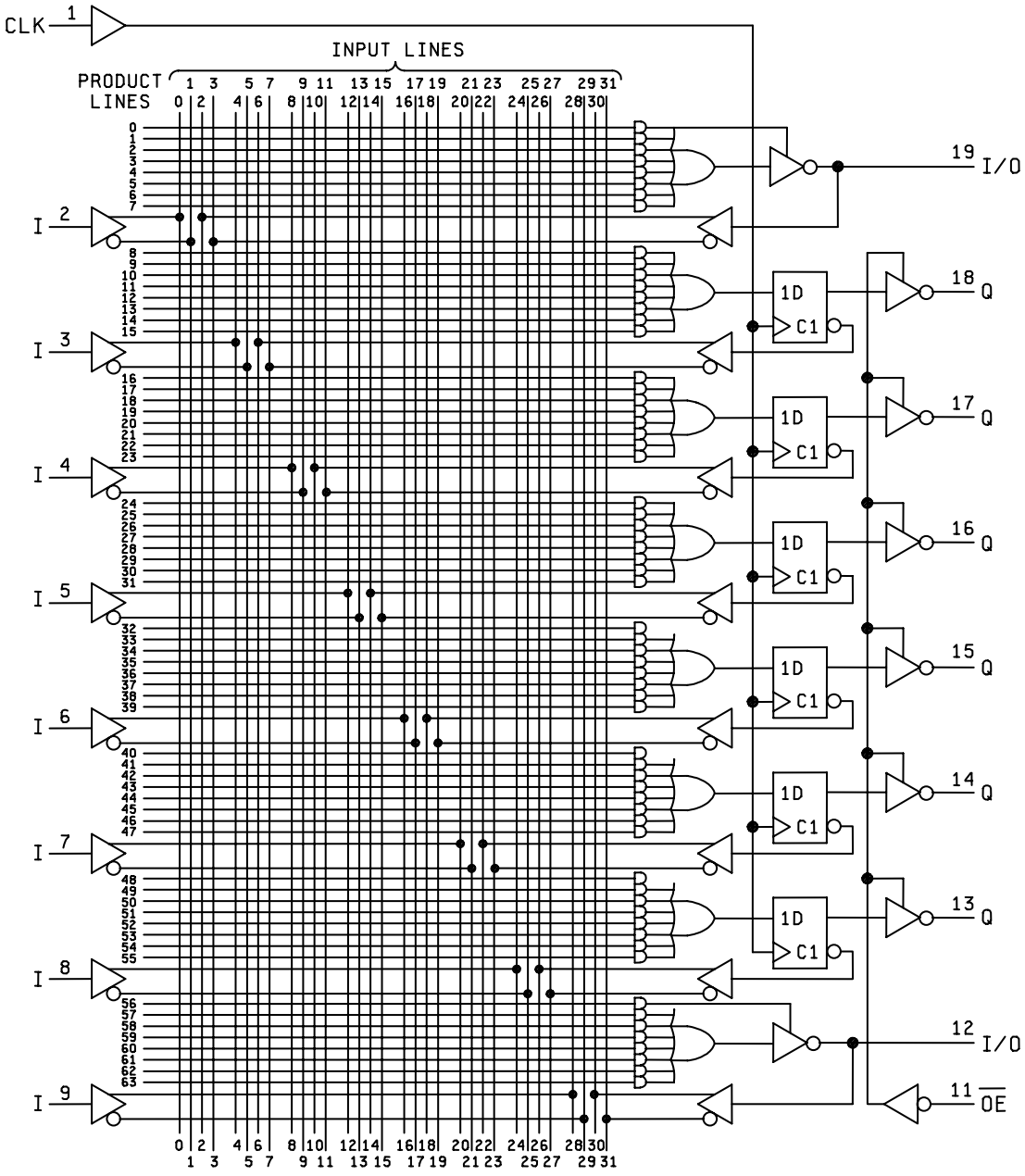



FIGURE 3. Logic diagrams (unprogrammed) – Continued.



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DEVICE TYPES 04,08 AND 12

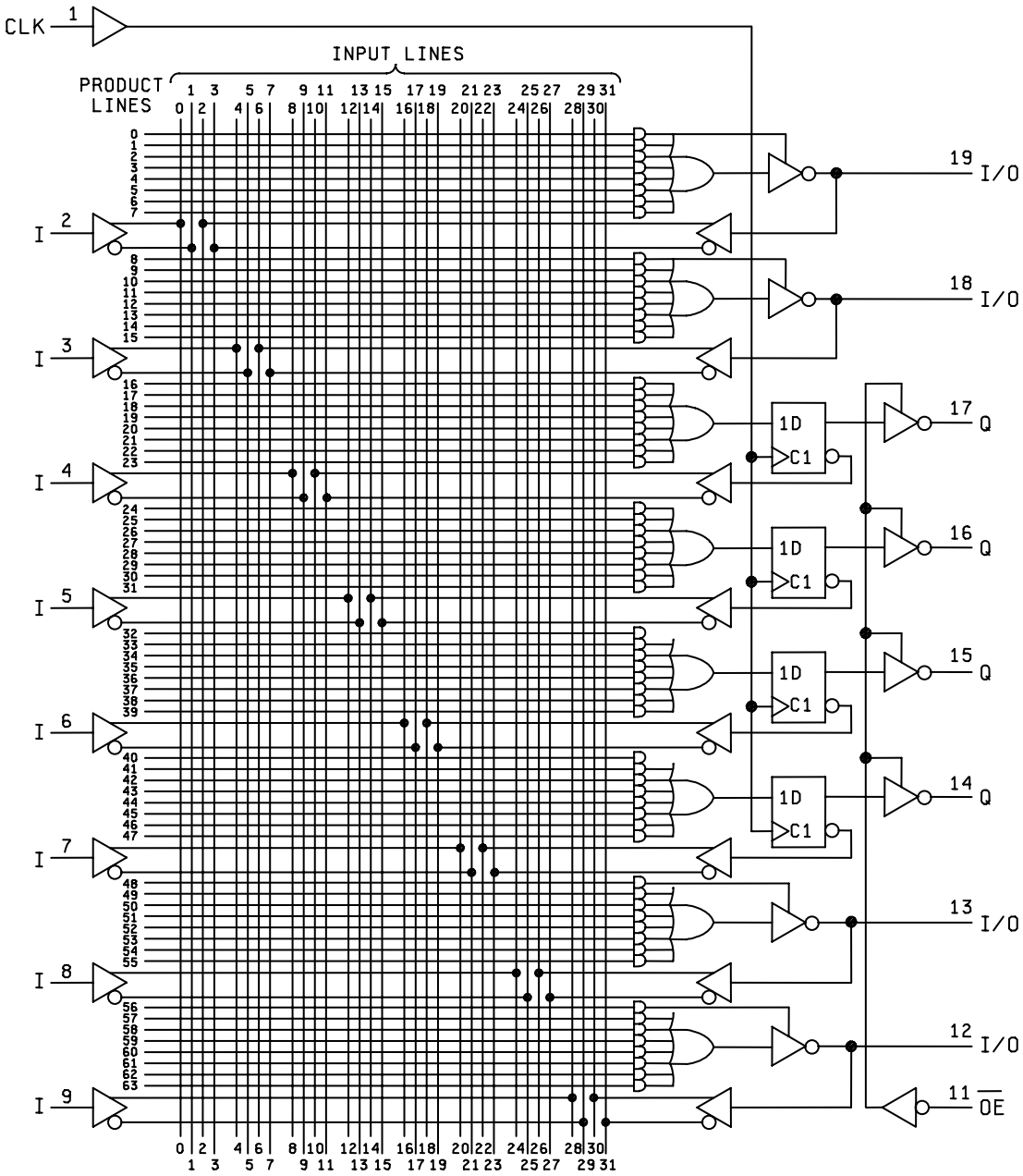
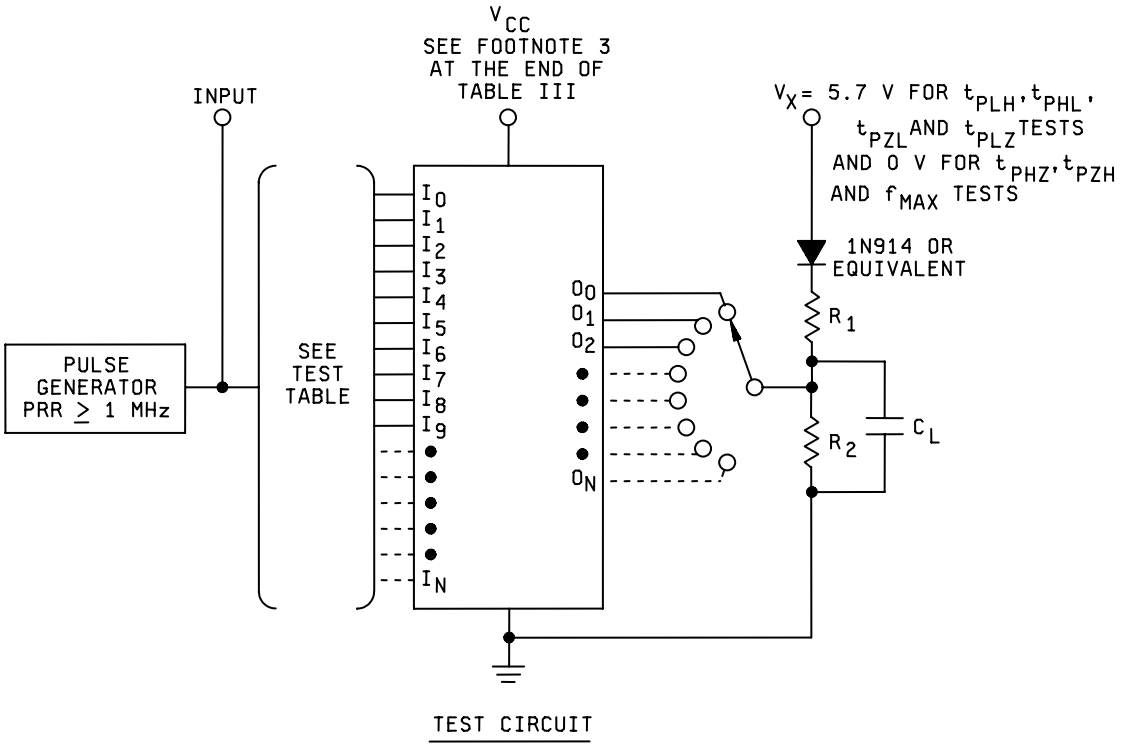


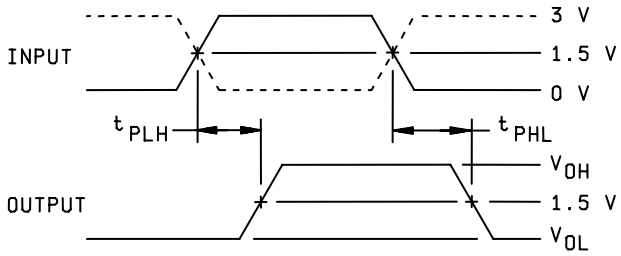

FIGURE 3. Logic diagrams (unprogrammed) – Continued.



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Check the source to verify that this is the current version before use.




TEST CIRCUIT

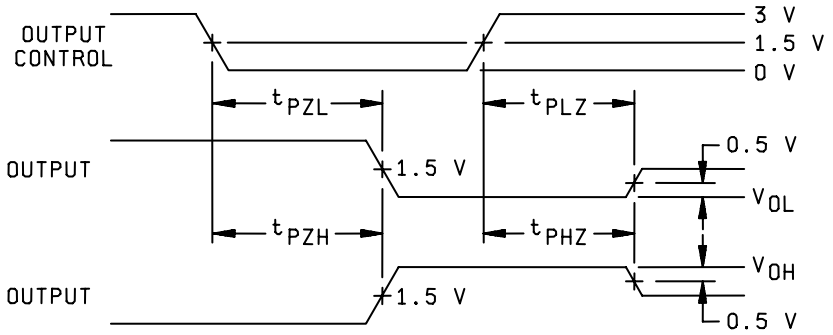



NON-REGISTERED SWITCHING WAVEFORMS

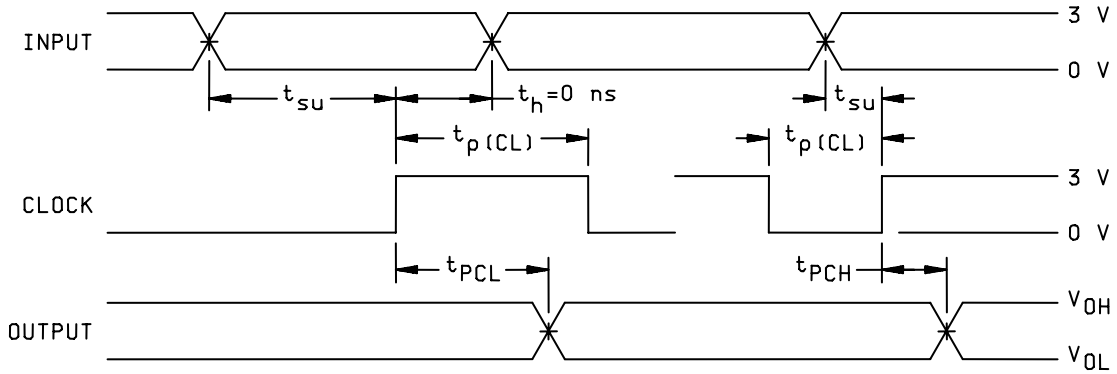
FIGURE 4. Switching waveforms.



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OUTPUT CONTROL SWITCHING WAVEFORM




REGISTERED SWITCHING WAVEFORM

NOTES:
1. Test table for devices programmed in accordance with an altered item drawing may be replaced by the equivalent tests which apply to the specific program configuration for the resulting PAL.

2. CL = 50 pF minimum, including jig and probe capacitance; R1 = 365Ω ±2%; R2 = 715Ω ±2%.

3. Ouputs may be under load simultaneously.

4. Verification of limit conditions $t_P$(CL). $t_{SU}$, and $t_h$ = 0 ns may be conducted by simultaneously setting the conditions to the specified limit values of table III and observing for proper output state change.

FIGURE 4. Switching waveforms – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:47Z
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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | " | | | | | $I_0$ | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $I_9$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | | -18 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | | -18 mA | $I/O_2$ | | " | | " | | " | " |
| | | 13 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 14 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 15 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 16 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 17 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ | 12 mA | | | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $I/O_2$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | | " | | " | | " | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | | " | | " | | " | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | | " | | " | | " | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 25 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I/O_1$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $I/O_2$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | " | | " | | " | | " |
| | | 29 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | " | | " | | " | | " |
| | | 30 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | " | | " | | " | | " |
| | | 31 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | | " | | " | | " |
| | | 32 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 33 | 0.4 V | | | | | | | | | " | | | | | $I_0$ | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 34 | | 0.4 V | | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 35 | | | 0.4 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 36 | | | | 0.4 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 37 | | | | | 0.4 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 38 | | | | | | 0.4 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 39 | | | | | | | 0.4 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 40 | | | | | | | | 0.4 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 41 | | | | | | | | | 0.4 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 42 | | | | | | | | | | " | 0.4 V | | | | $I_9$ | | " | | " | | " | " |
| | | 43 | | | | | | | | | | " | | | 0.4 V | | $I/O_1$ | | " | | " | | " | " |
| | | 44 | | | | | | | | | | " | | | | 0.4 V | $I/O_2$ | | " | | " | | " | " |
| | | 45 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 46 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 47 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 48 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | $I_0$ | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $I_9$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 12 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 13 | -18 mA | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 14 | | -18 mA | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 15 | | | -18 mA | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 16 | | | | -18 mA | | " | $I/O_6$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 17 | | | | | | " | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 18 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 19 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 20 | 12 mA | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 21 | | 12 mA | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 22 | | | 12 mA | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 23 | | | | 12 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 24 | | | | | 12 mA | " | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 25 | | | | | | " | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 26 | | | | | | " | $I/O_1$ | " | | " | | " | | " |
| | | 27 | | | | | | " | $I/O_2$ | " | | " | | " | | " |
| | | 28 | -2 mA | | | | | " | $I/O_3$ | " | | " | | " | | " |
| | | 29 | | -2 mA | | | | " | $I/O_4$ | " | | " | | " | | " |
| | | 30 | | | -2 mA | | | " | $I/O_5$ | " | | " | | " | | " |
| | | 31 | | | | -2 mA | | " | $I/O_6$ | " | | " | | " | | " |
| | | 32 | | | | | -2 mA | " | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 33 | | | | | | 5.5 V | $I_0$ | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 34 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I_9$ | | " | | " | | " | " |
| | | 43 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 45 | 0.4 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 46 | | 0.4 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 47 | | | 0.4 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 48 | | | | 0.4 V | | " | $I/O_6$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 49 | 2.7 V | | | | | | | | | “ | | | | | $I_0$ | | 50 | | 50 | | 50 | µA |
| | | 50 | | 2.7 V | | | | | | | | “ | | | | | $I_1$ | | 20 | | 20 | | 20 | “ |
| | | 51 | | | 2.7 V | | | | | | | “ | | | | | $I_2$ | | “ | | “ | | “ | “ |
| | | 52 | | | | 2.7 V | | | | | | “ | | | | | $I_3$ | | “ | | “ | | “ | “ |
| | | 53 | | | | | 2.7 V | | | | | “ | | | | | $I_4$ | | “ | | “ | | “ | “ |
| | | 54 | | | | | | 2.7 V | | | | “ | | | | | $I_5$ | | “ | | “ | | “ | “ |
| | | 55 | | | | | | | 2.7 V | | | “ | | | | | $I_6$ | | “ | | “ | | “ | “ |
| | | 56 | | | | | | | | 2.7 V | | “ | | | | | $I_7$ | | “ | | “ | | “ | “ |
| | | 57 | | | | | | | | | 2.7 V | “ | | | | | $I_8$ | | “ | | “ | | “ | “ |
| | | 58 | | | | | | | | | | “ | 2.7 V | | | | $I_9$ | | 50 | | 50 | | 50 | “ |
| $I_I$ | | 59 | 5.5 V | | | | | | | | | “ | | | | | $I_0$ | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 60 | | 5.5 V | | | | | | | | “ | | | | | $I_1$ | | 0.1 | | 0.1 | | 0.1 | “ |
| | | 61 | | | 5.5 V | | | | | | | “ | | | | | $I_2$ | | “ | | “ | | “ | “ |
| | | 62 | | | | 5.5 V | | | | | | “ | | | | | $I_3$ | | “ | | “ | | “ | “ |
| | | 63 | | | | | 5.5 V | | | | | “ | | | | | $I_4$ | | “ | | “ | | “ | “ |
| | | 64 | | | | | | 5.5 V | | | | “ | | | | | $I_5$ | | “ | | “ | | “ | “ |
| | | 65 | | | | | | | 5.5 V | | | “ | | | | | $I_6$ | | “ | | “ | | “ | “ |
| | | 66 | | | | | | | | 5.5 V | | “ | | | | | $I_7$ | | “ | | “ | | “ | “ |
| | | 67 | | | | | | | | | 5.5 V | “ | | | | | $I_8$ | | “ | | “ | | “ | “ |
| | | 68 | | | | | | | | | | “ | 5.5 V | | | | $I_9$ | | 0.2 | | 0.2 | | 0.2 | “ |
| | | 69 | | | | | | | | | | “ | | | 5.5 V | | $I/O_1$ | | 0.1 | | 0.1 | | 0.1 | “ |
| | | 70 | | | | | | | | | | “ | | | | 5.5 V | $I/O_2$ | | “ | | “ | | “ | “ |
| | | 71 | | | | | | | | | | “ | | | | | $I/O_3$ | | “ | | “ | | “ | “ |
| | | 72 | | | | | | | | | | “ | | | | | $I/O_4$ | | “ | | “ | | “ | “ |
| | | 73 | | | | | | | | | | “ | | | | | $I/O_5$ | | “ | | “ | | “ | “ |
| | | 74 | | | | | | | | | | “ | | | | | $I/O_6$ | | “ | | “ | | “ | “ |
| $I_O$ 11/ | | 75 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | “ | 1/ | 2.25 V | | | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | “ |
| | | 76 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | 2.25 V | | $I/O_1$ | “ | “ | “ | “ | “ | “ | “ |
| | | 77 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | 2.25 V | $I/O_2$ | “ | “ | “ | “ | “ | “ | “ |
| | | 78 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_3$ | “ | “ | “ | “ | “ | “ | “ |
| | | 79 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_4$ | “ | “ | “ | “ | “ | “ | “ |
| | | 80 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_5$ | “ | “ | “ | “ | “ | “ | “ |
| | | 81 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_6$ | “ | “ | “ | “ | “ | “ | “ |
| | | 82 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $O_7$ | “ | “ | “ | “ | “ | “ | “ |
| $I_{OZH}$ | | 83 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | 2.7 V | | | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 84 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | 2.7 V | | $I/O_1$ | | 100 | | 100 | | 100 | “ |
| | | 85 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | 2.7 V | $I/O_2$ | | “ | | “ | | “ | “ |
| | | 86 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_3$ | | “ | | “ | | “ | “ |
| | | 87 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_4$ | | “ | | “ | | “ | “ |
| | | 88 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_5$ | | “ | | “ | | “ | “ |
| | | 89 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $I/O_6$ | | “ | | “ | | “ | “ |
| | | 90 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $O_7$ | | 20 | | 20 | | 20 | “ |
| $I_{OZL}$ | | 91 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | 0.4 V | | | $O_0$ | | -20 | | -20 | | -20 | “ |
| | | 92 | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | “ | | | | $O_7$ | | -20 | | -20 | | -20 | “ |
| $I_{CC}$ | 3005 | 93 | | | | | | | | | | “ | | | | | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 01 - Continued

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 49 | | | | | | 5.5 V | $I_0$ | | 50 | | 50 | | 50 | µA |
| | | 50 | | | | | | " | $I_1$ | | 20 | | 20 | | 20 | " |
| | | 51 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 57 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_9$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 59 | | | | | | " | $I_0$ | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 60 | | | | | | " | $I_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 61 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 62 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 63 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 64 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 65 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 66 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 67 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 68 | | | | | | " | $I_9$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 69 | | | | | | " | $I/O_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 70 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 71 | 5.5 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 72 | | 5.5 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 73 | | | 5.5 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 74 | | | | 5.5 V | | " | $I/O_6$ | | " | | " | | " | " |
| $I_O$ 11/ | | 75 | | | | | | " | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 76 | | | | | | " | $I/O_1$ | " | " | " | " | " | " | " |
| | | 77 | | | | | | " | $I/O_2$ | " | " | " | " | " | " | " |
| | | 78 | 2.25 V | | | | | " | $I/O_3$ | " | " | " | " | " | " | " |
| | | 79 | | 2.25 V | | | | " | $I/O_4$ | " | " | " | " | " | " | " |
| | | 80 | | | 2.25 V | | | " | $I/O_5$ | " | " | " | " | " | " | " |
| | | 81 | | | | 2.25 V | | " | $I/O_6$ | " | " | " | " | " | " | " |
| | | 82 | | | | | 2.25 V | " | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 83 | | | | | | " | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 84 | | | | | | " | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 85 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 86 | 2.7 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 87 | | 2.7 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 88 | | | 2.7 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 89 | | | | 2.7 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 90 | | | | | 2.7 V | " | $O_7$ | | 20 | | 20 | | 20 | " |
| $I_{OZL}$ | | 91 | | | | | | " | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 92 | | | | | 0.4 V | " | $O_7$ | | -20 | | -20 | | -20 | " |
| $I_{CC}$ | 3005 | 93 | | | | | | " | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | I/$O_1$ | I/$O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 94 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{PHL}$ | 3003 | 95 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 3/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | All outputs | | 20 | | 20 | | 20 | ns |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Functional tests | 3014 | 94 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{PHL}$ | 3003 | 95 | 4/ 5/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | 3/ | All outputs | | 20 | | 20 | | 20 | ns |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 11 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 12 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $O_1$ | | " | | " | | " | " |
| | | 13 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 14 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 15 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 19 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $O_1$ | " | | " | | " | | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 27 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 28 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 29 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 30 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 31 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 32 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 33 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 34 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 35 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 36 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |





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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 11 | | | | | | " | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 12 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 13 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 14 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 15 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 16 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 17 | | | | 12 mA | | " | $O_6$ | | " | | " | | " | " |
| | | 18 | | | | | 12 mA | " | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 19 | | | | | | " | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 20 | | | | | | " | $O_1$ | " | | " | | " | | " |
| | | 21 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 22 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 23 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 24 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 25 | | | | -2 mA | | " | $O_6$ | " | | " | | " | | " |
| | | 26 | | | | | -2 mA | " | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 27 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 28 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 29 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 30 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 31 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 32 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 02 – Continued.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 37 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 38 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 39 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 40 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 41 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 42 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 43 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 44 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 45 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 46 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 47 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 48 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 49 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 50 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 51 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 52 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 53 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 54 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 55 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 56 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| $I_O$ 11/ | | 57 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 58 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $O_1$ | " | " | " | " | " | " | " |
| | | 59 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 60 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 61 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 62 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 63 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | " | " | " | " | " | " |
| | | 64 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 65 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 66 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $O_1$ | | " | | " | | " | " |
| | | 67 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | " | | " | | " | " |
| | | 68 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 69 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $I_{OZL}$ | | 73 | " | " | " | " | " | " | " | " | " | " | " | 0.4 V | | | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | 0.4 V | | $O_1$ | | " | | " | | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | " | | " | | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 81 | | | | | | | | | | " | | | | | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 37 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 38 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 39 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 43 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 45 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 46 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 47 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 48 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 49 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 51 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| $I_O$ 11/ | | 57 | | | | | | " | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 58 | | | | | | " | $O_1$ | " | " | " | " | " | " | " |
| | | 59 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 60 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 61 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 62 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 63 | | | | 2.25 V | | " | $O_6$ | " | " | " | " | " | " | " |
| | | 64 | | | | | 2.25 V | " | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 65 | | | | | | " | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 66 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 67 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 68 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 69 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 70 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 71 | | | | 2.7 V | | " | $O_6$ | | " | | " | | " | " |
| | | 72 | | | | | 2.7 V | " | $O_7$ | | " | | " | | " | " |
| $I_{OZL}$ | | 73 | | | | | | " | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 74 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 75 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 76 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 77 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 78 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 79 | | | | 0.4 V | | " | $O_6$ | | " | | " | | " | " |
| | | 80 | | | | | 0.4 V | " | $O_7$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 81 | | | | | | " | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table



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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 82 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 83 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | All 3/ register outputs | | 20 | | 20 | | 20 | ns |
| $t_{P(CL)}$ 10/ | | 84 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | 12 | | 12 | | 12 | | " |
| $t_{PCH}$ | | 85 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | 15 | | 15 | | 15 | " |
| $t_{PCL}$ | | 86 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 87 | " | " | " | " | " | " | " | " | " | " | 3/ | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 88 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 89 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 90 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 91 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | 9/ | 9/ | 9/ | " | 40 | | 40 | | 40 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 82 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 83 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | All 3/ register outputs | | 20 | | 20 | | 20 | ns |
| $t_{P(CL)}$ 10/ | | 84 | " | " | " | " | " | " | " | 12 | | 12 | | 12 | | " |
| $t_{PCH}$ | | 85 | " | " | " | " | " | " | " | | 15 | | 15 | | 15 | " |
| $t_{PCL}$ | | 86 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 87 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 88 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 89 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 90 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 91 | 9/ | 9/ | 9/ | 9/ | 9/ | 4.5 V | " | 40 | | 40 | | 40 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | -18 mA | | | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 13 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 14 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $O_1$ | | " | | " | | " | " |
| | | 15 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 21 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $O_1$ | " | | " | | " | | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 29 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 30 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 31 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 32 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 33 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 34 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 35 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 36 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 37 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 38 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 39 | | | | | | | | | | " | | 0.4 V | | | $I/O_0$ | | " | | " | | " | " |
| | | 40 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | -18 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 13 | | | | | | " | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 14 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 15 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 16 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 17 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 18 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 19 | | | | 12 mA | | " | $O_6$ | | " | | " | | " | " |
| | | 20 | | | | | 12 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 21 | | | | | | " | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 22 | | | | | | " | $O_1$ | " | | " | | " | | " |
| | | 23 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 24 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 25 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 26 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 27 | | | | -2 mA | | " | $O_6$ | " | | " | | " | | " |
| | | 28 | | | | | -2 mA | " | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 29 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 30 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 31 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 32 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 40 | | | | | 0.4 V | " | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 41 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 42 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 43 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 44 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 45 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 46 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 47 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 48 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 49 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 50 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 51 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 52 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 53 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 54 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 55 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 56 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 57 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 58 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 59 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 60 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 61 | | | | | | | | | | " | | 5.5 V | | | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 62 | | | | | | | | | | " | | | | | $I/O_7$ | | 0.1 | | 0.1 | | 0.1 | " |
| $I_O$ 11/ | | 63 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 64 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $O_1$ | " | " | " | " | " | " | " |
| | | 65 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 66 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 67 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 68 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 69 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | " | " | " | " | " | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 71 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $O_1$ | | 20 | | 20 | | 20 | " |
| | | 73 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | " | | " | | " | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 79 | " | " | " | " | " | " | " | " | " | " | " | | 0.4 V | | $O_1$ | | -20 | | -20 | | -20 | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | " | | " | | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 83 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 85 | | | | | | | | | | " | | | | | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 41 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 42 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 43 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 45 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 46 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 47 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 48 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 49 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 51 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 52 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 53 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 57 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 59 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 60 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 61 | | | | | | " | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 62 | | | | | 5.5 V | " | $I/O_7$ | | 0.1 | | 0.1 | | 0.1 | " |
| $I_O$ 11/ | | 63 | | | | | | " | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 64 | | | | | | " | $O_1$ | " | " | " | " | " | " | " |
| | | 65 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 66 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 67 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 68 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 69 | | | | 2.25 V | | " | $O_6$ | " | " | " | " | " | " | " |
| | | 70 | | | | | 2.25 V | " | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 71 | | | | | | " | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 72 | | | | | | " | $O_1$ | | 20 | | 20 | | 20 | " |
| | | 73 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 74 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 75 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 76 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 77 | | | | 2.7 V | | " | $O_6$ | | " | | " | | " | " |
| | | 78 | | | | | 2.7 V | " | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 79 | | | | | | " | $O_1$ | | -20 | | -20 | | -20 | " |
| | | 80 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 81 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 82 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 83 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 84 | | | | 0.4 V | | " | $O_6$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 85 | | | | | | " | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Functional tests | 3014 | 86 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 87 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | | 4/ 8/ | 4/ 8/ | All register outputs | 20 | | 20 | | 20 | | ns |
| $t_{P(CL)}$ 10/ | | 88 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | 12 | | 12 | | 12 | | " |
| $t_{PCL}$ | | 89 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | 15 | | 15 | | 15 | " |
| $t_{PCH}$ | | 90 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | 15 | | 15 | | 15 | " |
| $t_{PHL}$ | | 91 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PLH}$ | | 92 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZH}$ | | 93 | " | " | " | " | " | " | " | " | " | " | 3/ | | 4/ 8/ | 4/ 8/ | All register outputs | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 94 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 95 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 96 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | | " | | | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | | " | | | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | | " | | | " | | " | | " | | " | " |
| $f_{MAX}$ | | 101 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | | 9/ | 9/ | All register outputs | 40 | | 40 | | 40 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 86 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 87 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | | 3/ | All register outputs | 20 | | 20 | | 20 | | ns |
| $t_{P(CL)}$ 10/ | | 88 | " | " | " | " | | " | " | 12 | | 12 | | 12 | | " |
| $t_{PCL}$ | | 89 | " | " | " | " | | " | " | | 15 | | 15 | | 15 | " |
| $t_{PCH}$ | | 90 | " | " | " | " | | " | " | | 15 | | 15 | | 15 | " |
| $t_{PHL}$ | | 91 | | | | | 4/ 5 | " | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PLH}$ | | 92 | | | | | 4/ 5 | " | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZH}$ | | 93 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | | " | All register outputs | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 94 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 95 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 96 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | | | | | 4/ 5 | " | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZL}$ | | 98 | | | | | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | | | | | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | | | | | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 101 | 9/ | 9/ | 9/ | 9/ | | 4.5 V | All register outputs | 40 | | 40 | | 40 | | MHz |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | -18 mA | | | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | -18 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 13 | | | | | | | | | | | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 14 | | | | | | | | | | | | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 15 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 23 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I/O_1$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 29 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | | " | | " | | " |
| | | 30 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 31 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 32 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 33 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 34 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 35 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 36 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 37 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 38 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 39 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 40 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 41 | | | | | | | | | | " | | 0.4 V | | | $I/O_0$ | | " | | " | | " | " |
| | | 42 | | | | | | | | | | " | | | 0.4 V | | $I/O_1$ | | " | | " | | " | " |
| | | 43 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 44 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 13 | | | | -18 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 14 | | | | | -18 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 15 | | | | | | " | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 16 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 17 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 18 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 19 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 20 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 21 | | | | 12 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 22 | | | | | 12 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 23 | | | | | | " | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 24 | | | | | | " | $I/O_1$ | " | | " | | " | | " |
| | | 25 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 26 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 27 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 28 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 29 | | | | -2 mA | | " | $I/O_6$ | " | | " | | " | | " |
| | | 30 | | | | | -2 mA | " | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 31 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 32 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 43 | | | | 0.4 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 44 | | | | | 0.4 V | " | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 04 – Continued..

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 45 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 46 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 47 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 48 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 49 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 50 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 51 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 52 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 53 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 54 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 55 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 56 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 57 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 58 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 59 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 60 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 61 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 62 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 63 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 64 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 65 | | | | | | | | | | " | | 5.5 V | | | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 66 | | | | | | | | | | " | | | 5.5 V | | $I/O_1$ | | " | | " | | " | " |
| | | 67 | | | | | | | | | | | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 68 | | | | | | | | | | | | | | | $I/O_7$ | | " | | " | | " | " |
| $I_O$ 11/ | | 69 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $I/O_1$ | " | " | " | " | " | " | " |
| | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 73 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | " | " | " | " | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 77 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | 20 | | 20 | | 20 | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 83 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | 100 | | 100 | | 100 | " |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 85 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | -20 | | -20 | | -20 | " |
| | | 86 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 87 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 88 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 89 | | | | | | | | | | " | | | | | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 04 – Continued.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 45 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 46 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 47 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 48 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 49 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 51 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 55 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 56 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 57 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 59 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 60 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 61 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 62 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 63 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 64 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 65 | | | | | | " | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 66 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 67 | | | | 5.5 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 68 | | | | | 5.5 V | " | $I/O_7$ | | " | | " | | " | " |
| $I_O$ 11/ | | 69 | | | | | | " | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 70 | | | | | | " | $I/O_1$ | " | " | " | " | " | " | " |
| | | 71 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 72 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 73 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 74 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 75 | | | | 2.25 V | | " | $I/O_6$ | " | " | " | " | " | " | " |
| | | 76 | | | | | 2.25 V | " | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 77 | | | | | | " | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 78 | | | | | | " | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 79 | | | | | | " | $O_2$ | | 20 | | 20 | | 20 | " |
| | | 80 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 81 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 82 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 83 | | | | 2.7 V | | " | $I/O_6$ | | 100 | | 100 | | 100 | " |
| | | 84 | | | | | 2.7 V | " | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 85 | | | | | | " | $O_2$ | | -20 | | -20 | | -20 | " |
| | | 86 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 87 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 88 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 89 | | | | | | " | $V_{CC}$ | | 190 | | 190 | | 190 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_1$ | $I/O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Functional tests | 3014 | 90 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 91 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | | | 4/ 8/ | All register outputs | 20 | | 20 | | 20 | | ns |
| $t_{P(CL)}$ 10/ | | 92 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | 12 | | 12 | | 12 | | " |
| $t_{PCL}$ | | 93 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | 15 | | 15 | | 15 | " |
| $t_{PCH}$ | | 94 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | 15 | | 15 | | 15 | " |
| $t_{PHL}$ | | 95 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | 3/ | | | 4/ 8/ | All register outputs | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 101 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZL}$ | | 102 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 103 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 104 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $f_{MAX}$ | | 105 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | | | 9/ | All register outputs | 40 | | 40 | | 40 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 90 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 91 | 4/ 8/ | 4/ 8/ | 4/ 8/ | | | 3/ | All register outputs | 20 | | 20 | | 20 | | ns |
| $t_{P(CL)}$ 10/ | | 92 | " | " | " | | | " | " | 12 | | 12 | | 12 | | " |
| $t_{PCL}$ | | 93 | " | " | " | | | " | " | | 15 | | 15 | | 15 | " |
| $t_{PCH}$ | | 94 | " | " | " | | | " | " | | 15 | | 15 | | 15 | " |
| $t_{PHL}$ | | 95 | | | | 4/ 5 | 4/ 5/ | " | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PLH}$ | | 96 | | | | 4/ 5 | 4/ 5/ | " | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZH}$ | | 97 | 4/ 8/ | 4/ 8/ | 4/ 8/ | | | " | All register outputs | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 98 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 101 | | | | 4/ 5/ | 4/ 5 | " | All I/O's | | 20 | | 20 | | 20 | " |
| $t_{PZL}$ | | 102 | | | | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 103 | | | | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 104 | | | | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 105 | 9/ | 9/ | 9/ | | | 4.5 V | All register outputs | 40 | | 40 | | 40 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | $I_0$ | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $I_9$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | | -18 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | | -18 mA | $I/O_2$ | | " | | " | | " | " |
| | | 13 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 14 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 15 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 16 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 17 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ | 12 mA | | | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $I/O_2$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | | " | | " | | " | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | | " | | " | | " | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | | " | | " | | " | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 25 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I/O_1$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $I/O_2$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | " | | " | | " | | " |
| | | 29 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | " | | " | | " | | " |
| | | 30 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | " | | " | | " | | " |
| | | 31 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | | " | | " | | " |
| | | 32 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 33 | 0.4 V | | | | | | | | | " | | | | | $I_0$ | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 34 | | 0.4 V | | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 35 | | | 0.4 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 36 | | | | 0.4 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 37 | | | | | 0.4 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 38 | | | | | | 0.4 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 39 | | | | | | | 0.4 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 40 | | | | | | | | 0.4 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 41 | | | | | | | | | 0.4 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 42 | | | | | | | | | | " | 0.4 V | | | | $I_9$ | | " | | " | | " | " |
| | | 43 | | | | | | | | | | " | | | 0.4 V | | $I/O_1$ | | " | | " | | " | " |
| | | 44 | | | | | | | | | | " | | | | 0.4 V | $I/O_2$ | | " | | " | | " | " |
| | | 45 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 46 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 47 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 48 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 05– Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | $I_0$ | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $I_9$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 12 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 13 | -18 mA | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 14 | | -18 mA | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 15 | | | -18 mA | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 16 | | | | -18 mA | | " | $I/O_6$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 17 | | | | | | " | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 18 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 19 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 20 | 12 mA | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 21 | | 12 mA | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 22 | | | 12 mA | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 23 | | | | 12 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 24 | | | | | 12 mA | " | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 25 | | | | | | " | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 26 | | | | | | " | $I/O_1$ | " | | " | | " | | " |
| | | 27 | | | | | | " | $I/O_2$ | " | | " | | " | | " |
| | | 28 | -2 mA | | | | | " | $I/O_3$ | " | | " | | " | | " |
| | | 29 | | -2 mA | | | | " | $I/O_4$ | " | | " | | " | | " |
| | | 30 | | | -2 mA | | | " | $I/O_5$ | " | | " | | " | | " |
| | | 31 | | | | -2 mA | | " | $I/O_6$ | " | | " | | " | | " |
| | | 32 | | | | | -2 mA | " | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 33 | | | | | | 5.5 V | $I_0$ | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 34 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I_9$ | | " | | " | | " | " |
| | | 43 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 45 | 0.4 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 46 | | 0.4 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 47 | | | 0.4 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 48 | | | | 0.4 V | | " | $I/O_6$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 49 | 2.7 V | | | | | | | | | GND | | | | | $I_0$ | | 50 | | 50 | | 50 | µA |
| | | 50 | | 2.7 V | | | | | | | | " | | | | | $I_1$ | | 20 | | 20 | | 20 | " |
| | | 51 | | | 2.7 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 52 | | | | 2.7 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 53 | | | | | 2.7 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 54 | | | | | | 2.7 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 55 | | | | | | | 2.7 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 56 | | | | | | | | 2.7 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 57 | | | | | | | | | 2.7 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 58 | | | | | | | | | | " | 2.7 V | | | | $I_9$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 59 | 5.5 V | | | | | | | | | " | | | | | $I_0$ | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 60 | | 5.5 V | | | | | | | | " | | | | | $I_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 61 | | | 5.5 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 62 | | | | 5.5 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 63 | | | | | 5.5 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 64 | | | | | | 5.5 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 65 | | | | | | | 5.5 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 66 | | | | | | | | 5.5 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 67 | | | | | | | | | 5.5 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 68 | | | | | | | | | | " | 5.5 V | | | | $I_9$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 69 | | | | | | | | | | " | | | 5.5 V | | $I/O_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 70 | | | | | | | | | | " | | | | 5.5 V | $I/O_2$ | | " | | " | | " | " |
| | | 71 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 72 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 73 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 74 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| $I_O$ 11/ | | 75 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ | 2.25 V | | | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $I/O_1$ | " | " | " | " | " | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $I/O_2$ | " | " | " | " | " | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | " | " | " | " | " | " | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | " | " | " | " | " | " | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | " | " | " | " | " | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | " | " | " | " | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 83 | " | " | " | " | " | " | " | " | " | " | " | 2.7 V | | | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 85 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $I/O_2$ | | " | | " | | " | " |
| | | 86 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | | " | | " | | " | " |
| | | 87 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | | " | | " | | " | " |
| | | 88 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | | " | | " | | " | " |
| | | 89 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 90 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | 20 | | 20 | | 20 | " |
| $I_{OZL}$ | | 91 | " | " | " | " | " | " | " | " | " | " | " | 0.4 V | | | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 92 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | -20 | | -20 | | -20 | " |
| $I_{CC}$ | 3005 | 93 | | | | | | | | | | " | | | | | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 49 | | | | | | 5.5 V | $I_0$ | | 50 | | 50 | | 50 | µA |
| | | 50 | | | | | | " | $I_1$ | | 20 | | 20 | | 20 | " |
| | | 51 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 57 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_9$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 59 | | | | | | " | $I_0$ | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 60 | | | | | | " | $I_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 61 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 62 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 63 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 64 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 65 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 66 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 67 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 68 | | | | | | " | $I_9$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 69 | | | | | | " | $I/O_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 70 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 71 | 5.5 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 72 | | 5.5 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 73 | | | 5.5 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 74 | | | | 5.5 V | | " | $I/O_6$ | | " | | " | | " | " |
| $I_O$ 11/ | | 75 | | | | | | " | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 76 | | | | | | " | $I/O_1$ | " | " | " | " | " | " | " |
| | | 77 | | | | | | " | $I/O_2$ | " | " | " | " | " | " | " |
| | | 78 | 2.25 V | | | | | " | $I/O_3$ | " | " | " | " | " | " | " |
| | | 79 | | 2.25 V | | | | " | $I/O_4$ | " | " | " | " | " | " | " |
| | | 80 | | | 2.25 V | | | " | $I/O_5$ | " | " | " | " | " | " | " |
| | | 81 | | | | 2.25 V | | " | $I/O_6$ | " | " | " | " | " | " | " |
| | | 82 | | | | | 2.25 V | " | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 83 | | | | | | " | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 84 | | | | | | " | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 85 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 86 | 2.7 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 87 | | 2.7 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 88 | | | 2.7 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 89 | | | | 2.7 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 90 | | | | | 2.7 V | " | $O_7$ | | 20 | | 20 | | 20 | " |
| $I_{OZL}$ | | 91 | | | | | | " | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 92 | | | | | 0.4 V | " | $O_7$ | | -20 | | -20 | | -20 | " |
| $I_{CC}$ | | 93 | | | | | | " | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 94 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{PHL}$ | 3003 | 95 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 3/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | All outputs | | 30 | | 30 | | 30 | ns |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 94 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{PHL}$ | 3003 | 95 | 4/ 5/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | 3/ | All outputs | | 30 | | 30 | | 30 | ns |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | " | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 11 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 12 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $O_1$ | | " | | " | | " | " |
| | | 13 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 14 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 15 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 19 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $O_1$ | " | | " | | " | | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 27 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 28 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 29 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 30 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 31 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 32 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 33 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 34 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 35 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 36 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 11 | | | | | | " | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 12 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 13 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 14 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 15 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 16 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 17 | | | | 12 mA | | " | $O_6$ | | " | | " | | " | " |
| | | 18 | | | | | 12 mA | " | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 19 | | | | | | " | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 20 | | | | | | " | $O_1$ | " | | " | | " | | " |
| | | 21 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 22 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 23 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 24 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 25 | | | | -2 mA | | " | $O_6$ | " | | " | | " | | " |
| | | 26 | | | | | -2 mA | " | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 27 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 28 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 29 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 30 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 31 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 32 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 37 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 38 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 39 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 40 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 41 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 42 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 43 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 44 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 45 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 46 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 47 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 48 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 49 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 50 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 51 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 52 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 53 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 54 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 55 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 56 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| $I_O$ 11/ | | 57 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 58 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $O_1$ | " | " | " | " | " | " | " |
| | | 59 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 60 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 61 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 62 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 63 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | " | " | " | " | " | " |
| | | 64 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 65 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 66 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $O_1$ | | " | | " | | " | " |
| | | 67 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | " | | " | | " | " |
| | | 68 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 69 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $I_{OZL}$ | | 73 | " | " | " | " | " | " | " | " | " | " | " | 0.4 V | | | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | 0.4 V | | $O_1$ | | " | | " | | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | " | | " | | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 81 | | | | | | | | | | " | | | | | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 37 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 38 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 39 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 43 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 45 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 46 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 47 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 48 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 49 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 51 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| $I_O$ 11/ | | 57 | | | | | | " | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 58 | | | | | | " | $O_1$ | " | " | " | " | " | " | " |
| | | 59 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 60 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 61 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 62 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 63 | | | | 2.25 V | | " | $O_6$ | " | " | " | " | " | " | " |
| | | 64 | | | | | 2.25 V | " | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 65 | | | | | | " | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 66 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 67 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 68 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 69 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 70 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 71 | | | | 2.7 V | | " | $O_6$ | | " | | " | | " | " |
| | | 72 | | | | | 2.7 V | " | $O_7$ | | " | | " | | " | " |
| $I_{OZL}$ | | 73 | | | | | | " | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 74 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 75 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 76 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 77 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 78 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 79 | | | | 0.4 V | | " | $O_6$ | | " | | " | | " | " |
| | | 80 | | | | | 0.4 V | " | $O_7$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 81 | | | | | | " | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 82 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 83 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | All register outputs | 25 | | 25 | | 25 | | ns |
| $t_{P(CL)}$ 10/ | | 84 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | 20 | | 20 | | 20 | | " |
| $t_{PCH}$ | | 85 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | 20 | | 20 | | 20 | " |
| $t_{PCL}$ | | 86 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | 20 | | 20 | | 20 | " |
| $t_{PZH}$ | | 87 | " | " | " | " | " | " | " | " | " | " | 3/ | " | " | " | " | | 25 | | 25 | | 25 | " |
| $t_{PZL}$ | | 88 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 89 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 90 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 91 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | 9/ | 9/ | 9/ | " | 25 | | 25 | | 25 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 82 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 83 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | 3/ | All register outputs | 25 | | 25 | | 25 | | ns |
| $t_{P(CL)}$ 10/ | | 84 | " | " | " | " | " | " | " | 20 | | 20 | | 20 | | " |
| $t_{PCH}$ | | 85 | " | " | " | " | " | " | " | | 20 | | 20 | | 20 | " |
| $t_{PCL}$ | | 86 | " | " | " | " | " | " | " | | 20 | | 20 | | 20 | " |
| $t_{PZH}$ | | 87 | " | " | " | " | " | " | " | | 25 | | 25 | | 25 | " |
| $t_{PZL}$ | | 88 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 89 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 90 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 91 | 9/ | 9/ | 9/ | 9/ | 9/ | 4.5 V | " | 25 | | 25 | | 25 | | MHz |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 07.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | -18 mA | | | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 13 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 14 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $O_1$ | | " | | " | | " | " |
| | | 15 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 21 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $O_1$ | " | | " | | " | | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | | | | | $O_6$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | | | | | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 29 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 30 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 31 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 32 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 33 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 34 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 35 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 36 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 37 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 38 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 39 | | | | | | | | | | " | | 0.4 V | | | $I/O_0$ | | " | | " | | " | " |
| | | 40 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | -18 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 13 | | | | | | " | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 14 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 15 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 16 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 17 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 18 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 19 | | | | 12 mA | | " | $O_6$ | | " | | " | | " | " |
| | | 20 | | | | | 12 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 21 | | | | | | " | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 22 | | | | | | " | $O_1$ | " | | " | | " | | " |
| | | 23 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 24 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 25 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 26 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 27 | | | | -2 mA | | " | $O_6$ | " | | " | | " | | " |
| | | 28 | | | | | -2 mA | " | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 29 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 30 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 31 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 32 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 40 | | | | | 0.4 V | " | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 41 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 42 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 43 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 44 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 45 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 46 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 47 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 48 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 49 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 50 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 51 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 52 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 53 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 54 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 55 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 56 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 57 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 58 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 59 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 60 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 61 | | | | | | | | | | " | | 5.5 V | | | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 62 | | | | | | | | | | " | | | | | $I/O_7$ | | 0.1 | | 0.1 | | 0.1 | " |
| $I_O$ 11/ | | 63 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 64 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $O_1$ | " | " | " | " | " | " | " |
| | | 65 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 66 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 67 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 68 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 69 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | " | " | " | " | " | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 71 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $O_1$ | | 20 | | 20 | | 20 | " |
| | | 73 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | " | | " | | " | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | | | | | | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_1$ | | -20 | | -20 | | -20 | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | 0.4 V | | $O_2$ | | " | | " | | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_3$ | | " | | " | | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 83 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 85 | | | | | | | | | | " | | | | | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 41 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 42 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 43 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 45 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 46 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 47 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 48 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 49 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 51 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 52 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 53 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 57 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 59 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 60 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 61 | | | | | | " | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 62 | | | | | 5.5 V | " | $I/O_7$ | | 0.1 | | 0.1 | | 0.1 | " |
| $I_O$ 11/ | | 63 | | | | | | " | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 64 | | | | | | " | $O_1$ | " | " | " | " | " | " | " |
| | | 65 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 66 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 67 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 68 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 69 | | | | 2.25 V | | " | $O_6$ | " | " | " | " | " | " | " |
| | | 70 | | | | | 2.25 V | " | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 71 | | | | | | " | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 72 | | | | | | " | $O_1$ | | 20 | | 20 | | 20 | " |
| | | 73 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 74 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 75 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 76 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 77 | | | | 2.7 V | | " | $O_6$ | | | | | | | " |
| | | 78 | | | | | 2.7 V | " | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 79 | | | | | | " | $O_1$ | | -20 | | -20 | | -20 | " |
| | | 80 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 81 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 82 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 83 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 84 | | | | 0.4 V | | " | $O_6$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 85 | | | | | | " | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Functional tests | 3014 | 86 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 87 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | | 4/ 8/ | 4/ 8/ | All register outputs | 25 | | 25 | | 25 | | ns |
| $t_{P(CL)}$ 10/ | | 88 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | 20 | | 20 | | 20 | | " |
| $t_{PCL}$ | | 89 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | 20 | | 20 | | 20 | " |
| $t_{PCH}$ | | 90 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | 20 | | 20 | | 20 | " |
| $t_{PHL}$ | | 91 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PLH}$ | | 92 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZH}$ | | 93 | " | " | " | " | " | " | " | " | " | " | 3/ | | 4/ 8/ | 4/ 8/ | All register outputs | | 25 | | 25 | | 25 | " |
| $t_{PZL}$ | | 94 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 95 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 96 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | " | 4/ 5/ | | | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | " | | | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | " | | | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | " | | | " | | " | | " | | " | " |
| $f_{MAX}$ | | 101 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | | 9/ | 9/ | All register outputs | 25 | | 25 | | 25 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 86 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 87 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | | 3/ | All register outputs | 25 | | 25 | | 25 | | ns |
| $t_{P(CL)}$ 10/ | | 88 | " | " | " | " | | " | " | 20 | | 20 | | 20 | | " |
| $t_{PCL}$ | | 89 | " | " | " | " | | " | " | | 20 | | 20 | | 20 | " |
| $t_{PCH}$ | | 90 | " | " | " | " | | " | " | | 20 | | 20 | | 20 | " |
| $t_{PHL}$ | | 91 | | | | | 4/ 5/ | " | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PLH}$ | | 92 | | | | | 4/ 5/ | " | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZH}$ | | 93 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | | " | All register outputs | | 25 | | 25 | | 25 | " |
| $t_{PZL}$ | | 94 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 95 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 96 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | | | | | 4/ 5/ | " | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZL}$ | | 98 | | | | | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | | | | | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | | | | | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 101 | 9/ | 9/ | 9/ | 9/ | | 4.5 V | All register outputs | 25 | | 25 | | 25 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 08.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | -18 mA | | | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | -18 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 13 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 14 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 15 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 23 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I/O_1$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 29 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | | " | | " | | " |
| | | 30 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 31 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 32 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 33 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 34 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 35 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 36 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 37 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 38 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 39 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 40 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 41 | | | | | | | | | | " | | 0.4 V | | | $I/O_0$ | | " | | " | | " | " |
| | | 42 | | | | | | | | | | " | | | 0.4 V | | $I/O_1$ | | " | | " | | " | " |
| | | 43 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 44 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 13 | | | | -18 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 14 | | | | | -18 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 15 | | | | | | " | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 16 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 17 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 18 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 19 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 20 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 21 | | | | 12 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 22 | | | | | 12 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 23 | | | | | | " | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 24 | | | | | | " | $I/O_1$ | " | | " | | " | | " |
| | | 25 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 26 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 27 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 28 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 29 | | | | -2 mA | | " | $I/O_6$ | " | | " | | " | | " |
| | | 30 | | | | | -2 mA | " | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 31 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 32 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 43 | | | | 0.4 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 44 | | | | | 0.4 V | " | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 45 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 46 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 47 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 48 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 49 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 50 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 51 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 52 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 53 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 54 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 55 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 56 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 57 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 58 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 59 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 60 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 61 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 62 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 63 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 64 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 65 | | | | | | | | | | " | | 5.5 V | | | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 66 | | | | | | | | | | " | | | 5.5 V | | $I/O_1$ | | " | | " | | " | " |
| | | 67 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 68 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |
| $I_O$ 11/ | | 69 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $I/O_1$ | " | " | " | " | " | " | " |
| | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 73 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | " | " | " | " | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 77 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | 20 | | 20 | | 20 | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 83 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | 100 | | 100 | | 100 | " |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 85 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | -20 | | -20 | | -20 | " |
| | | 86 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 87 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 88 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 89 | | | | | | | | | | " | | | | | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 45 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 46 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 47 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 48 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 49 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 51 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 55 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 56 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 57 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 59 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 60 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 61 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 62 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 63 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 64 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 65 | | | | | | " | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 66 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 67 | | | | 5.5 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 68 | | | | | 5.5 V | " | $I/O_7$ | | " | | " | | " | " |
| $I_O$ 11/ | | 69 | | | | | | " | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 70 | | | | | | " | $I/O_1$ | " | " | " | " | " | " | " |
| | | 71 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 72 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 73 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 74 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 75 | | | | 2.25 V | | " | $I/O_6$ | " | " | " | " | " | " | " |
| | | 76 | | | | | 2.25 V | " | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 77 | | | | | | " | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 78 | | | | | | " | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 79 | | | | | | " | $O_2$ | | 20 | | 20 | | 20 | " |
| | | 80 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 81 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 82 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 83 | | | | 2.7 V | | " | $I/O_6$ | | 100 | | 100 | | 100 | " |
| | | 84 | | | | | 2.7 V | " | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 85 | | | | | | " | $O_2$ | | -20 | | -20 | | -20 | " |
| | | 86 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 87 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 88 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 89 | | | | | | " | $V_{CC}$ | | 105 | | 105 | | 105 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Functional tests | 3014 | 90 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 91 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | | | 4/ 8/ | All register outputs | 25 | | 25 | | 25 | | ns |
| $t_{P(CL)}$ 10/ | | 92 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | 20 | | 20 | | 20 | | " |
| $t_{PCL}$ | | 93 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | 20 | | 20 | | 20 | " |
| $t_{PCH}$ | | 94 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | 20 | | 20 | | 20 | " |
| $t_{PHL}$ | | 95 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | 3/ | | | 4/ 8/ | All register outputs | | 25 | | 25 | | 25 | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 101 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZL}$ | | 102 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 103 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 104 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $f_{MAX}$ | | 105 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | | | 9/ | All register outputs | 25 | | 25 | | 25 | | MHz |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 90 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 91 | 4/ 8/ | 4/ 8/ | 4/ 8/ | | | 3/ | All register outputs | 25 | | 25 | | 25 | | ns |
| $t_{P(CL)}$ 10/ | | 92 | " | " | " | | | " | " | 20 | | 20 | | 20 | | " |
| $t_{PCL}$ | | 93 | " | " | " | | | " | " | | 20 | | 20 | | 20 | " |
| $t_{PCH}$ | | 94 | " | " | " | | | " | " | | 20 | | 20 | | 20 | " |
| $t_{PHL}$ | | 95 | | | | 4/ 5/ | 4/ 5/ | " | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PLH}$ | | 96 | | | | 4/ 5/ | 4/ 5/ | " | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZH}$ | | 97 | 4/ 8/ | 4/ 8/ | 4/ 8/ | | | " | All register outputs | | 25 | | 25 | | 25 | " |
| $t_{PZL}$ | | 98 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 101 | | | | 4/ 5/ | 4/ 5/ | " | All I/O's | | 30 | | 30 | | 30 | " |
| $t_{PZL}$ | | 102 | | | | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 103 | | | | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 104 | | | | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 105 | 9/ | 9/ | 9/ | | | 4.5 V | All register outputs | 25 | | 25 | | 25 | | MHz |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 09.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | $I_0$ | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $I_9$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | " | | | -18 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | " | | | | -18 mA | $I/O_2$ | | " | | " | | " | " |
| | | 13 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 14 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 15 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 16 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 17 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ | 12 mA | | | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $I/O_2$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | | " | | " | | " | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | | " | | " | | " | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | | " | | " | | " | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 25 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I/O_1$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $I/O_2$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | " | | " | | " | | " |
| | | 29 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | " | | " | | " | | " |
| | | 30 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | " | | " | | " | | " |
| | | 31 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | | " | | " | | " |
| | | 32 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 33 | 0.4 V | | | | | | | | | " | | | | | $I_0$ | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 34 | | 0.4 V | | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 35 | | | 0.4 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 36 | | | | 0.4 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 37 | | | | | 0.4 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 38 | | | | | | 0.4 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 39 | | | | | | | 0.4 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 40 | | | | | | | | 0.4 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 41 | | | | | | | | | 0.4 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 42 | | | | | | | | | | " | 0.4 V | | | | $I_9$ | | " | | " | | " | " |
| | | 43 | | | | | | | | | | " | | | 0.4 V | | $I/O_1$ | | " | | " | | " | " |
| | | 44 | | | | | | | | | | " | | | | 0.4 V | $I/O_2$ | | " | | " | | " | " |
| | | 45 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 46 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 47 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 48 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 09 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | $I_0$ | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $I_9$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 12 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 13 | -18 mA | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 14 | | -18 mA | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 15 | | | -18 mA | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 16 | | | | -18 mA | | " | $I/O_6$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 17 | | | | | | " | $O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 18 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 19 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 20 | 12 mA | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 21 | | 12 mA | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 22 | | | 12 mA | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 23 | | | | 12 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 24 | | | | | 12 mA | " | $O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 25 | | | | | | " | $O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 26 | | | | | | " | $I/O_1$ | " | | " | | " | | " |
| | | 27 | | | | | | " | $I/O_2$ | " | | " | | " | | " |
| | | 28 | -2 mA | | | | | " | $I/O_3$ | " | | " | | " | | " |
| | | 29 | | -2 mA | | | | " | $I/O_4$ | " | | " | | " | | " |
| | | 30 | | | -2 mA | | | " | $I/O_5$ | " | | " | | " | | " |
| | | 31 | | | | -2 mA | | " | $I/O_6$ | " | | " | | " | | " |
| | | 32 | | | | | -2 mA | " | $O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 33 | | | | | | 5.5 V | $I_0$ | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 34 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I_9$ | | " | | " | | " | " |
| | | 43 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 45 | 0.4 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 46 | | 0.4 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 47 | | | 0.4 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 48 | | | | 0.4 V | | " | $I/O_6$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 09 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 49 | 2.7 V | | | | | | | | | GND | | | | | $I_0$ | | 50 | | 50 | | 50 | µA |
| | | 50 | | 2.7 V | | | | | | | | " | | | | | $I_1$ | | 20 | | 20 | | 20 | " |
| | | 51 | | | 2.7 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 52 | | | | 2.7 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 53 | | | | | 2.7 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 54 | | | | | | 2.7 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 55 | | | | | | | 2.7 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 56 | | | | | | | | 2.7 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 57 | | | | | | | | | 2.7 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 58 | | | | | | | | | | " | 2.7 V | | | | $I_9$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 59 | 5.5 V | | | | | | | | | " | | | | | $I_0$ | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 60 | | 5.5 V | | | | | | | | " | | | | | $I_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 61 | | | 5.5 V | | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 62 | | | | 5.5 V | | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 63 | | | | | 5.5 V | | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 64 | | | | | | 5.5 V | | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 65 | | | | | | | 5.5 V | | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 66 | | | | | | | | 5.5 V | | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 67 | | | | | | | | | 5.5 V | " | | | | | $I_8$ | | " | | " | | " | " |
| | | 68 | | | | | | | | | | " | 5.5 V | | | | $I_9$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 69 | | | | | | | | | | " | | | 5.5 V | | $I/O_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 70 | | | | | | | | | | " | | | | 5.5 V | $I/O_2$ | | " | | " | | " | " |
| | | 71 | | | | | | | | | | " | | | | | $I/O_3$ | | " | | " | | " | " |
| | | 72 | | | | | | | | | | " | | | | | $I/O_4$ | | " | | " | | " | " |
| | | 73 | | | | | | | | | | " | | | | | $I/O_5$ | | " | | " | | " | " |
| | | 74 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| $I_O$ 11/ | | 75 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ | 2.25 V | | | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $I/O_1$ | " | " | " | " | " | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $I/O_2$ | " | " | " | " | " | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | " | " | " | " | " | " | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | " | " | " | " | " | " | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | " | " | " | " | " | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | " | " | " | " | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | " | | " | | " | " |
| $I_{OZH}$ | | 83 | " | " | " | " | " | " | " | " | " | " | " | 2.7 V | | | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 85 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $I/O_2$ | | " | | " | | " | " |
| | | 86 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_3$ | | " | | " | | " | " |
| | | 87 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_4$ | | " | | " | | " | " |
| | | 88 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_5$ | | " | | " | | " | " |
| | | 89 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 90 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | 20 | | 20 | | 20 | " |
| $I_{OZL}$ | | 91 | " | " | " | " | " | " | " | " | " | " | " | 0.4 V | | | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 92 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | -20 | | -20 | | -20 | " |
| $I_{CC}$ | 3005 | 93 | | | | | | | | | | " | | | | | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 09 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 49 | | | | | | 5.5 V | $I_0$ | | 50 | | 50 | | 50 | µA |
| | | 50 | | | | | | " | $I_1$ | | 20 | | 20 | | 20 | " |
| | | 51 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 57 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_9$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 59 | | | | | | " | $I_0$ | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 60 | | | | | | " | $I_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 61 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 62 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 63 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 64 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 65 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 66 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 67 | | | | | | " | $I_8$ | | " | | " | | " | " |
| | | 68 | | | | | | " | $I_9$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 69 | | | | | | " | $I/O_1$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 70 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 71 | 5.5 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 72 | | 5.5 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 73 | | | 5.5 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 74 | | | | 5.5 V | | " | $I/O_6$ | | " | | " | | " | " |
| $I_O$ 11/ | | 75 | | | | | | " | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 76 | | | | | | " | $I/O_1$ | " | " | " | " | " | " | " |
| | | 77 | | | | | | " | $I/O_2$ | " | " | " | " | " | " | " |
| | | 78 | 2.25 V | | | | | " | $I/O_3$ | " | " | " | " | " | " | " |
| | | 79 | | 2.25 V | | | | " | $I/O_4$ | " | " | " | " | " | " | " |
| | | 80 | | | 2.25 V | | | " | $I/O_5$ | " | " | " | " | " | " | " |
| | | 81 | | | | 2.25 V | | " | $I/O_6$ | " | " | " | " | " | " | " |
| | | 82 | | | | | 2.25 V | " | $O_7$ | " | " | | " | | " | " |
| $I_{OZH}$ | | 83 | | | | | | " | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 84 | | | | | | " | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 85 | | | | | | " | $I/O_2$ | | " | | " | | " | " |
| | | 86 | 2.7 V | | | | | " | $I/O_3$ | | " | | " | | " | " |
| | | 87 | | 2.7 V | | | | " | $I/O_4$ | | " | | " | | " | " |
| | | 88 | | | 2.7 V | | | " | $I/O_5$ | | " | | " | | " | " |
| | | 89 | | | | 2.7 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 90 | | | | | 2.7 V | " | $O_7$ | | 20 | | 20 | | 20 | " |
| $I_{OZL}$ | | 91 | | | | | | " | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 92 | | | | | 0.4 V | " | $O_7$ | | -20 | | -20 | | -20 | " |
| $I_{CC}$ | 3005 | 93 | | | | | | " | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

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TABLE III. Group A inspection for device type 09 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | $I_8$ | GND | $I_9$ | $O_0$ | $I/O_1$ | $I/O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 94 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{PHL}$ | 3003 | 95 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 3/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | All outputs | | 15 | | 15 | | 15 | ns |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 09 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $I/O_3$ | $I/O_4$ | $I/O_5$ | $I/O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 94 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{PHL}$ | 3003 | 95 | 4/ 5/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | 4/ 5/ | 3/ | All outputs | | 15 | | 15 | | 15 | ns |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 10.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 11 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 12 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I_1$ | | " | | " | | " | " |
| | | 13 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $I_2$ | | " | | " | | " | " |
| | | 14 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_3$ | | " | | " | | " | " |
| | | 15 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_4$ | | " | | " | | " | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_5$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_6$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 19 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I_1$ | " | | " | | " | | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $I_2$ | " | | " | | " | | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_3$ | " | | " | | " | | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_4$ | " | | " | | " | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_5$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_6$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $I_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 27 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 28 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 29 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 30 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 31 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 32 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 33 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 34 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 35 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 36 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 11 | | | | | | " | $I_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 12 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 13 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 14 | 12 mA | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 15 | | 12 mA | | | | " | $I_4$ | | " | | " | | " | " |
| | | 16 | | | 12 mA | | | " | $I_5$ | | " | | " | | " | " |
| | | 17 | | | | 12 mA | | " | $I_6$ | | " | | " | | " | " |
| | | 18 | | | | | 12 mA | " | $I_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 19 | | | | | | " | $I_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 20 | | | | | | " | $I_1$ | " | | " | | " | | " |
| | | 21 | | | | | | " | $I_2$ | " | | " | | " | | " |
| | | 22 | -2 mA | | | | | " | $I_3$ | " | | " | | " | | " |
| | | 23 | | -2 mA | | | | " | $I_4$ | " | | " | | " | | " |
| | | 24 | | | -2 mA | | | " | $I_5$ | " | | " | | " | | " |
| | | 25 | | | | -2 mA | | " | $I_6$ | " | | " | | " | | " |
| | | 26 | | | | | -2 mA | " | $I_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 27 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 28 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 29 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 30 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 31 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 32 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 37 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 38 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 39 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 40 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 41 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 42 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 43 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 44 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 45 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 46 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 47 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 48 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 49 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 50 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 51 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 52 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 53 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 54 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 55 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 56 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| $I_O$ 11/ | | 57 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 58 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $O_1$ | " | " | " | " | " | " | " |
| | | 59 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 60 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 61 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 62 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 63 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | " | " | " | " | " | " |
| | | 64 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 65 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 66 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $O_1$ | | " | | " | | " | " |
| | | 67 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | " | | " | | " | " |
| | | 68 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 69 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $I_{OZL}$ | | 73 | " | " | " | " | " | " | " | " | " | " | " | 0.4 V | | | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | 0.4 V | | $O_1$ | | " | | " | | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | " | | " | | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_7$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 81 | | | | | | | | | | " | | | | | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 37 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 38 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 39 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 43 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 45 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 46 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 47 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 48 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 49 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 51 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| $I_O$ 11/ | | 57 | | | | | | " | $O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 58 | | | | | | " | $O_1$ | " | " | " | " | " | " | " |
| | | 59 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 60 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 61 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 62 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 63 | | | | 2.25 V | | " | $O_6$ | " | " | " | " | " | " | " |
| | | 64 | | | | | 2.25 V | " | $O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 65 | | | | | | " | $O_0$ | | 20 | | 20 | | 20 | µA |
| | | 66 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 67 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 68 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 69 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 70 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 71 | | | | 2.7 V | | " | $O_6$ | | " | | " | | " | " |
| | | 72 | | | | | 2.7 V | " | $O_7$ | | " | | " | | " | " |
| $I_{OZL}$ | | 73 | | | | | | " | $O_0$ | | -20 | | -20 | | -20 | " |
| | | 74 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 75 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 76 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 77 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 78 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 79 | | | | 0.4 V | | " | $O_6$ | | " | | " | | " | " |
| | | 80 | | | | | 0.4 V | " | $O_7$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 81 | | | | | | " | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $O_0$ | $O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 82 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 83 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | All register outputs | 15 | | 15 | | 15 | | ns |
| $t_{P(CL)}$ 10/ | | 84 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | 10 | | 10 | | 10 | | " |
| $t_{PCH}$ | | 85 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | 12 | | 12 | | 12 | " |
| $t_{PCL}$ | | 86 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 87 | " | " | " | " | " | " | " | " | " | " | 3/ | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 88 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 89 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 90 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 91 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | 9/ | 9/ | 9/ | " | 50 | | 50 | | 50 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 82 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 83 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | 3/ | All register outputs | 15 | | 15 | | 15 | | ns |
| $t_{P(CL)}$ 10/ | | 84 | " | " | " | " | " | " | " | 10 | | 10 | | 10 | | " |
| $t_{PCH}$ | | 85 | " | " | " | " | " | " | " | | 12 | | 12 | | 12 | " |
| $t_{PCL}$ | | 86 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 87 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PZL}$ | | 88 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 89 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 90 | " | " | " | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 91 | 9/ | 9/ | 9/ | 9/ | 9/ | 4.5 V | " | 50 | | 50 | | 50 | | MHz |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 11.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | | | -18 mA | | | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | | | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 13 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 14 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $O_1$ | | " | | " | | " | " |
| | | 15 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 21 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $O_1$ | " | | " | | " | | " |
| | | 23 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 29 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 30 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 31 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 32 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 33 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 34 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 35 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 36 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 37 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 38 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 39 | | | | | | | | | | " | | 0.4 V | | | $I/O_0$ | | " | | " | | " | " |
| | | 40 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | -18 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 13 | | | | | | " | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 14 | | | | | | " | $O_1$ | | " | | " | | " | " |
| | | 15 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 16 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 17 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 18 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 19 | | | | 12 mA | | " | $O_6$ | | " | | " | | " | " |
| | | 20 | | | | | 12 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 21 | | | | | | " | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 22 | | | | | | " | $O_1$ | " | | " | | " | | " |
| | | 23 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 24 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 25 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 26 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 27 | | | | -2 mA | | " | $O_6$ | " | | " | | " | | " |
| | | 28 | | | | | -2 mA | " | $I/O_7$ | " | | " | | " | | " |
| $I_{IL}$ | 3009 | 29 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 30 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 31 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 32 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 40 | | | | | 0.4 V | " | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 41 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 42 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 43 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 44 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 45 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 46 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 47 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 48 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 49 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 50 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 51 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 52 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 53 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 54 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 55 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 56 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 57 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 58 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 59 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 60 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 61 | | | | | | | | | | " | | 5.5 V | | | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 62 | | | | | | | | | | " | | | | | $I/O_7$ | | 0.1 | | 0.1 | | 0.1 | " |
| $I_O$ 11/ | | 63 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $I/O_0$ | -15 | -125 | -158 | -125 | -15 | -125 | " |
| | | 64 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $O_1$ | " | " | " | " | " | " | " |
| | | 65 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 66 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 67 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 68 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 69 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $O_6$ | " | " | " | " | " | " | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_1$ | | 20 | | 20 | | 20 | " |
| | | 73 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_2$ | | " | | " | | " | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 77 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 79 | " | " | " | " | " | " | " | " | " | " | " | | 0.4 V | | $O_1$ | | -20 | | -20 | | -20 | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | " | | " | | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 83 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_6$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 85 | | | | | | | | | | " | | | | | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 41 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 42 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 43 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 44 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 45 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 46 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 47 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 48 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 49 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 51 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 52 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 53 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 55 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 56 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 57 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 59 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 60 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 61 | | | | | | " | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 62 | | | | | 5.5 V | " | $I/O_7$ | | 0.1 | | 0.1 | | 0.1 | " |
| $I_O$ 11/ | | 63 | | | | | | " | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 64 | | | | | | " | $O_1$ | " | " | " | " | " | " | " |
| | | 65 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 66 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 67 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 68 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 69 | | | | 2.25 V | | " | $O_6$ | " | " | " | " | " | " | " |
| | | 70 | | | | | 2.25 V | " | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 71 | | | | | | " | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 72 | | | | | | " | $O_1$ | | 20 | | 20 | | 20 | " |
| | | 73 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 74 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 75 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 76 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 77 | | | | 2.7 V | | " | $O_6$ | | " | | " | | " | " |
| | | 78 | | | | | 2.7 V | " | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 79 | | | | | | " | $O_1$ | | -20 | | -20 | | -20 | " |
| | | 80 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 81 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 82 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 83 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 84 | | | | 0.4 V | | " | $O_6$ | | " | | " | | " | " |
| $I_{CC}$ | | 85 | | | | | | " | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Functional tests | 3014 | 86 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 87 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | | 4/ 8/ | 4/ 8/ | All register outputs | 15 | | 15 | | 15 | | ns |
| $t_{P(CL)}$ 10/ | | 88 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | 10 | | 10 | | 10 | | " |
| $t_{PCL}$ | | 89 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | 12 | | 12 | | 12 | " |
| $t_{PCH}$ | | 90 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | 12 | | 12 | | 12 | " |
| $t_{PHL}$ | | 91 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PLH}$ | | 92 | " | " | " | " | " | " | " | " | " | " | | 4/ 5 | | | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZH}$ | | 93 | " | " | " | " | " | " | " | " | " | " | 3/ | | 4/ 8/ | 4/ 8/ | All register outputs | | 12 | | 12 | | 12 | " |
| $t_{PZL}$ | | 94 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 95 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 96 | " | " | " | " | " | " | " | " | " | " | " | | " | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | | | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | | " | | | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | | " | | | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | | " | | | " | | " | | " | | " | " |
| $f_{MAX}$ | | 101 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | | 9/ | 9/ | All register outputs | 50 | | 50 | | 50 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 86 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 87 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | | 3/ | All register outputs | 15 | | 15 | | 15 | | ns |
| $t_{P(CL)}$ 10/ | | 88 | " | " | " | " | | " | " | 10 | | 10 | | 10 | | " |
| $t_{PCL}$ | | 89 | " | " | " | " | | " | " | | 12 | | 12 | | 12 | " |
| $t_{PCH}$ | | 90 | " | " | " | " | | " | " | | 12 | | 12 | | 12 | " |
| $t_{PHL}$ | | 91 | | | | | 4/ 5 | " | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PLH}$ | | 92 | | | | | 4/ 5 | " | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZH}$ | | 93 | 4/ 8/ | 4/ 8/ | 4/ 8/ | 4/ 8/ | | " | All register outputs | | 12 | | 12 | | 12 | " |
| $t_{PZL}$ | | 94 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 95 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 96 | " | " | " | " | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 97 | | | | | 4/ 5 | " | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 98 | | | | | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | | | | | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | | | | | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 101 | 9/ | 9/ | 9/ | 9/ | | 4.5 V | All register outputs | 50 | | 50 | | 50 | | MHz |

See footnotes at end of table.



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TABLE III. Group A inspection for device type 12.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | -18 mA | | | | | | | | | GND | | | | | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | -18 mA | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 3 | | | -18 mA | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | -18 mA | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | -18 mA | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | -18 mA | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | | -18 mA | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | | | -18 mA | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | | | | -18 mA | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | | | | | " | -18 mA | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | | | | | | | -18 mA | | | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | | | | | | | | -18 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 13 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 14 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 15 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 12 mA | | | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 16 | " | " | " | " | " | " | " | " | " | " | " | | 12 mA | | $I/O_1$ | | " | | " | | " | " |
| | | 17 | " | " | " | " | " | " | " | " | " | " | " | | | 12 mA | $O_2$ | | " | | " | | " | " |
| | | 18 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 19 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 20 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 21 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | " | | " | | " | " |
| | | 22 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 23 | " | " | " | " | " | " | " | " | " | " | " | -2 mA | | | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 24 | " | " | " | " | " | " | " | " | " | " | " | | -2 mA | | $I/O_1$ | " | | " | | " | | " |
| | | 25 | " | " | " | " | " | " | " | " | " | " | " | | | -2 mA | $O_2$ | " | | " | | " | | " |
| | | 26 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | | " | | " | | " |
| | | 27 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | | " | | " | | " |
| | | 28 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | | " | | " | | " |
| | | 29 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | | " | | " | | |
| | | 30 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | | " | | " | | |
| $I_{IL}$ | 3009 | 31 | 0.4 V | | | | | | | | | " | | | | | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 32 | | 0.4 V | | | | | | | | " | | | | | $I_0$ | | " | | " | | " | " |
| | | 33 | | | 0.4 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 34 | | | | 0.4 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 35 | | | | | 0.4 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 36 | | | | | | 0.4 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 37 | | | | | | | 0.4 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 38 | | | | | | | | 0.4 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 39 | | | | | | | | | 0.4 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 40 | | | | | | | | | | " | 0.4 V | | | | $\overline{OE}$ | | " | | " | | " | " |
| | | 41 | | | | | | | | | | " | | 0.4 V | | | $I/O_0$ | | " | | " | | " | " |
| | | 42 | | | | | | | | | | " | | | 0.4 V | | $I/O_1$ | | " | | " | | " | " |
| | | 43 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 44 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 12 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $V_{IC}$ | | 1 | | | | | | 4.5 V | CLOCK | | -1.5 | | -1.5 | | -1.5 | V |
| | | 2 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 3 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 4 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 5 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 6 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 7 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 8 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 9 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 10 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 11 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 12 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 13 | | | | -18 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 14 | | | | | -18 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OL}$ | 3007 | 15 | | | | | | " | $I/O_0$ | | 0.5 | | 0.5 | | 0.5 | " |
| | | 16 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 17 | | | | | | " | $O_2$ | | " | | " | | " | " |
| | | 18 | 12 mA | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 19 | | 12 mA | | | | " | $O_4$ | | " | | " | | " | " |
| | | 20 | | | 12 mA | | | " | $O_5$ | | " | | " | | " | " |
| | | 21 | | | | 12 mA | | " | $I/O_6$ | | " | | " | | " | " |
| | | 22 | | | | | 12 mA | " | $I/O_7$ | | " | | " | | " | " |
| $V_{OH}$ | 3006 | 23 | | | | | | " | $I/O_0$ | 2.4 | | 2.4 | | 2.4 | | " |
| | | 24 | | | | | | " | $I/O_1$ | " | | " | | " | | " |
| | | 25 | | | | | | " | $O_2$ | " | | " | | " | | " |
| | | 26 | -2 mA | | | | | " | $O_3$ | " | | " | | " | | " |
| | | 27 | | -2 mA | | | | " | $O_4$ | " | | " | | " | | " |
| | | 28 | | | -2 mA | | | " | $O_5$ | " | | " | | " | | " |
| | | 29 | | | | -2 mA | | " | $I/O_6$ | " | | " | | " | | |
| | | 30 | | | | | -2 mA | " | $I/O_7$ | " | | " | | " | | |
| $I_{IL}$ | 3009 | 31 | | | | | | 5.5 V | CLOCK | | -0.25 | | -0.25 | | -0.25 | mA |
| | | 32 | | | | | | " | $I_0$ | | " | | " | | " | " |
| | | 33 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 34 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 35 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 36 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 37 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 38 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 39 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 40 | | | | | | " | $\overline{OE}$ | | " | | " | | " | " |
| | | 41 | | | | | | " | $I/O_0$ | | " | | " | | " | " |
| | | 42 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 43 | | | | 0.4 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 44 | | | | | 0.4 V | " | $I/O_7$ | | " | | " | | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 12 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 45 | 2.7 V | | | | | | | | | GND | | | | | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 46 | | 2.7 V | | | | | | | | " | | | | | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 47 | | | 2.7 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 48 | | | | 2.7 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 49 | | | | | 2.7 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 50 | | | | | | 2.7 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 51 | | | | | | | 2.7 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 52 | | | | | | | | 2.7 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 53 | | | | | | | | | 2.7 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 54 | | | | | | | | | | " | 2.7 V | | | | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 55 | 5.5 V | | | | | | | | | " | | | | | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 56 | | 5.5 V | | | | | | | | " | | | | | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 57 | | | 5.5 V | | | | | | | " | | | | | $I_1$ | | " | | " | | " | " |
| | | 58 | | | | 5.5 V | | | | | | " | | | | | $I_2$ | | " | | " | | " | " |
| | | 59 | | | | | 5.5 V | | | | | " | | | | | $I_3$ | | " | | " | | " | " |
| | | 60 | | | | | | 5.5 V | | | | " | | | | | $I_4$ | | " | | " | | " | " |
| | | 61 | | | | | | | 5.5 V | | | " | | | | | $I_5$ | | " | | " | | " | " |
| | | 62 | | | | | | | | 5.5 V | | " | | | | | $I_6$ | | " | | " | | " | " |
| | | 63 | | | | | | | | | 5.5 V | " | | | | | $I_7$ | | " | | " | | " | " |
| | | 64 | | | | | | | | | | " | 5.5 V | | | | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 65 | | | | | | | | | | " | | 5.5 V | | | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 66 | | | | | | | | | | " | | | 5.5 V | | $I/O_1$ | | " | | " | | " | " |
| | | 67 | | | | | | | | | | " | | | | | $I/O_6$ | | " | | " | | " | " |
| | | 68 | | | | | | | | | | " | | | | | $I/O_7$ | | " | | " | | " | " |
| $I_O$ 11/ | | 69 | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | 1/ | " | 1/ 6/ | 2.25 V | | | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 70 | " | " | " | " | " | " | " | " | " | " | " | | 2.25 V | | $I/O_1$ | " | " | " | " | " | " | " |
| | | 71 | " | " | " | " | " | " | " | " | " | " | " | | | 2.25 V | $O_2$ | " | " | " | " | " | " | " |
| | | 72 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | " | " | " | " | " | " | " |
| | | 73 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | " | " | " | " | " | " | " |
| | | 74 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | " | " | " | " | " | " | " |
| | | 75 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | " | " | " | " | " | " | " |
| | | 76 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 77 | " | " | " | " | " | " | " | " | " | " | 1/ 7/ | 2.7 V | | | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 78 | " | " | " | " | " | " | " | " | " | " | " | | 2.7 V | | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 79 | " | " | " | " | " | " | " | " | " | " | " | | | 2.7 V | $O_2$ | | 20 | | 20 | | 20 | " |
| | | 80 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 81 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 82 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| | | 83 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_6$ | | 100 | | 100 | | 100 | " |
| | | 84 | " | " | " | " | " | " | " | " | " | " | " | | | | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 85 | " | " | " | " | " | " | " | " | " | " | " | | | 0.4 V | $O_2$ | | -20 | | -20 | | -20 | " |
| | | 86 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_3$ | | " | | " | | " | " |
| | | 87 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_4$ | | " | | " | | " | " |
| | | 88 | " | " | " | " | " | " | " | " | " | " | " | | | | $O_5$ | | " | | " | | " | " |
| $I_{CC}$ | 3005 | 89 | | | | | | | | | | " | | | | | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 12 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Subgroup 1 $T_C$ = +25°C | | Subgroup 2 $T_C$ = +125°C | | Subgroup 3 $T_C$ = -55°C | | |
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $O_6$ | $I/O_7$ | $V_{CC}$ | | Min | Max | Min | Max | Min | Max | |
| $I_{IH}$ | 3010 | 45 | | | | | | 5.5 V | CLOCK | | 50 | | 50 | | 50 | µA |
| | | 46 | | | | | | " | $I_0$ | | 20 | | 20 | | 20 | " |
| | | 47 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 48 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 49 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 50 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 51 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 52 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 53 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 54 | | | | | | " | $\overline{OE}$ | | 50 | | 50 | | 50 | " |
| $I_I$ | | 55 | | | | | | " | CLOCK | | 0.2 | | 0.2 | | 0.2 | mA |
| | | 56 | | | | | | " | $I_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 57 | | | | | | " | $I_1$ | | " | | " | | " | " |
| | | 58 | | | | | | " | $I_2$ | | " | | " | | " | " |
| | | 59 | | | | | | " | $I_3$ | | " | | " | | " | " |
| | | 60 | | | | | | " | $I_4$ | | " | | " | | " | " |
| | | 61 | | | | | | " | $I_5$ | | " | | " | | " | " |
| | | 62 | | | | | | " | $I_6$ | | " | | " | | " | " |
| | | 63 | | | | | | " | $I_7$ | | " | | " | | " | " |
| | | 64 | | | | | | " | $\overline{OE}$ | | 0.2 | | 0.2 | | 0.2 | " |
| | | 65 | | | | | | " | $I/O_0$ | | 0.1 | | 0.1 | | 0.1 | " |
| | | 66 | | | | | | " | $I/O_1$ | | " | | " | | " | " |
| | | 67 | | | | 5.5 V | | " | $I/O_6$ | | " | | " | | " | " |
| | | 68 | | | | | 5.5 V | " | $I/O_7$ | | " | | " | | " | " |
| $I_O$ 11/ | | 69 | | | | | | " | $I/O_0$ | -15 | -125 | -15 | -125 | -15 | -125 | " |
| | | 70 | | | | | | " | $I/O_1$ | " | " | " | " | " | " | " |
| | | 71 | | | | | | " | $O_2$ | " | " | " | " | " | " | " |
| | | 72 | 2.25 V | | | | | " | $O_3$ | " | " | " | " | " | " | " |
| | | 73 | | 2.25 V | | | | " | $O_4$ | " | " | " | " | " | " | " |
| | | 74 | | | 2.25 V | | | " | $O_5$ | " | " | " | " | " | " | " |
| | | 75 | | | | 2.25 V | | " | $I/O_6$ | " | " | " | " | " | " | " |
| | | 76 | | | | | 2.25 V | " | $I/O_7$ | " | " | " | " | " | " | " |
| $I_{OZH}$ | | 77 | | | | | | " | $I/O_0$ | | 100 | | 100 | | 100 | µA |
| | | 78 | | | | | | " | $I/O_1$ | | 100 | | 100 | | 100 | " |
| | | 79 | | | | | | " | $O_2$ | | 20 | | 20 | | 20 | " |
| | | 80 | 2.7 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 81 | | 2.7 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 82 | | | 2.7 V | | | " | $O_5$ | | " | | " | | " | " |
| | | 83 | | | | 2.7 V | | " | $I/O_6$ | | 100 | | 100 | | 100 | " |
| | | 84 | | | | | 2.7 V | " | $I/O_7$ | | 100 | | 100 | | 100 | " |
| $I_{OZL}$ | | 85 | | | | | | " | $O_2$ | | -20 | | -20 | | -20 | " |
| | | 86 | 0.4 V | | | | | " | $O_3$ | | " | | " | | " | " |
| | | 87 | | 0.4 V | | | | " | $O_4$ | | " | | " | | " | " |
| | | 88 | | | 0.4 V | | | " | $O_5$ | | " | | " | | " | " |
| $I_{CC}$ | | 89 | | | | | | " | $V_{CC}$ | | 220 | | 220 | | 220 | mA |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 12 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Meas. terminal | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | CLK | $I_0$ | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ | GND | $\overline{OE}$ | $I/O_0$ | $I/O_1$ | $O_2$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 90 | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | GND | 2/ | 2/ | 2/ | 2/ | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 91 | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | GND | 6/ | | | 4/ 8/ | All register outputs | 15 | | 15 | | 15 | | ns |
| $t_{P(CL)}$ 10/ | | 92 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | 10 | | 10 | | 10 | | " |
| $t_{PCL}$ | | 93 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | 12 | | 12 | | 12 | " |
| $t_{PCH}$ | | 94 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | 12 | | 12 | | 12 | " |
| $t_{PHL}$ | | 95 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5/ | | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PLH}$ | | 96 | " | " | " | " | " | " | " | " | " | " | | 4/ 5 | 4/ 5 | | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZH}$ | | 97 | " | " | " | " | " | " | " | " | " | " | 3/ | | | 4/ 8/ | All register outputs | | 12 | | 12 | | 12 | " |
| $t_{PZL}$ | | 98 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | " | " | " | " | " | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 101 | " | " | " | " | " | " | " | " | " | " | | 4/ 5/ | 4/ 5 | | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 102 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 103 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 104 | " | " | " | " | " | " | " | " | " | " | | " | " | | " | | " | | " | | " | " |
| $f_{MAX}$ | | 105 | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | 9/ | " | 6/ | | | 9/ | All register outputs | 50 | | 50 | | 50 | | MHz |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 12 – Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

| Symbol | MIL-STD-883 method | Cases R, 2 | 15 | 16 | 17 | 18 | 19 | 20 | Meas. terminals | Test limits | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Test No. | $O_3$ | $O_4$ | $O_5$ | $I/O_6$ | $I/O_7$ | $V_{CC}$ | | Subgroup 7 $T_C$ = +25°C | | Subgroup 8 $T_C$ = +125°C | | Subgroup 8 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Func-tional tests | 3014 | 90 | 2/ | 2/ | 2/ | 2/ | 2/ | 5.5 V | All outputs | | 2/ | | 2/ | | 2/ | |
| | | | | | | | | | | Subgroup 9 $T_C$ = +25°C | | Subgroup 10 $T_C$ = +125°C | | Subgroup 11 $T_C$ = -55°C | | |
| | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| $t_{SU}$ 10/ | 3003 | 91 | 4/ 8/ | 4/ 8/ | 4/ 8/ | | | 3/ | All register outputs | 15 | | 15 | | 15 | | ns |
| $t_{P(CL)}$ 10/ | | 92 | " | " | " | " | " | " | " | 10 | | 10 | | 10 | | " |
| $t_{PCL}$ | | 93 | " | " | " | " | " | " | " | | 12 | | 12 | | 12 | " |
| $t_{PCH}$ | | 94 | " | " | " | " | " | " | " | | 12 | | 12 | | 12 | " |
| $t_{PHL}$ | | 95 | | | | 4/ 5 | 4/ 5 | " | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PLH}$ | | 96 | | | | 4/ 5 | 4/ 5 | " | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZH}$ | | 97 | 4/ 8/ | 4/ 8/ | 4/ 8/ | | | " | All register outputs | | 12 | | 12 | | 12 | " |
| $t_{PZL}$ | | 98 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 99 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 100 | " | " | " | | | " | " | | " | | " | | " | " |
| $t_{PZH}$ | | 101 | | | | 4/ 5 | 4/ 5 | " | All I/O's | | 15 | | 15 | | 15 | " |
| $t_{PZL}$ | | 102 | | | | " | " | " | " | | " | | " | | " | " |
| $t_{PHZ}$ | | 103 | | | | " | " | " | " | | " | | " | | " | " |
| $t_{PLZ}$ | | 104 | | | | " | " | " | " | | " | | " | | " | " |
| $f_{MAX}$ | | 105 | 9/ | 9/ | 9/ | | | 4.5 V | All register outputs | 50 | | 50 | | 50 | | MHz |

See footnotes at end of table.





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1/ For unprogrammed devices circuit C, for $V_{OL}$, $V_{OH}$, $I_{OZH}$, $I_{OZL}$, and $I_{OS}$, use conditions dfined for the programming procedures as follows:

$V_{OL}$:
a. 10.5 V on pins 2-9, GND on pins 10 and 11, and a clock pulse on pin 1. Outputs are on pins 13, 15, 17, 19.
b. 10.5 V on pins 3-9, GND 2, 10, and 11, and a clock pulse on pin 1. Outputs are on pins 12 and 14.
c. 10.5 V onn pins 3-9, GND on pins 1, 2, 10, and 11. Outputs are on pins 16 and 18.

$V_{OH}$:
a. 10.5 V on pin 11, GND on pins 2-10, and a clock pulse on pin 1. Outputs are on pins 12 - 15.
b. 10.5 V on pin 1, GND on pins 2-10, and a clock pulse on pin 11. Outputs are on pins 16 - 19.

$I_{OZH}$:
a. GND on pins 1-11, outputs are on pins 12-19 for device types 01 and 07. Outputs are on pins 12, 19 for device tyupes 03 and 09. Outputs are on pins 12, 13, 18, and 19 for device types 04 and 10.
b. GND on pins 1-10, 2.0 V on pin 11, outputs are on pins 12-19 for device types 02 and 08, 13-18 for device types 03 and 09. Outputs are on pins 14-17 for device types 04 and 10.

$I_{OZL}$:
a. GND on pins 2-11, and a clock pulse on pin 1. Outputs are on pins 12 and 19 for device types 01 and 07.
b. GND on pins 2-10, 2.0 V on pin 11, and a clock pulse on pin 1. Outputs are on pins 12-19 for device types 02 and 08. Ouputs are on pins 13-18 for device types 03 and 09. Outputs are on pins 14-17 for device types 04 and 10.

$I_{OS}$:
Applying the proper output test, generate the output condition as specified for $V_{OH}$. Short only one output at a time. Duration of the short should not exceed 1 second.

2/ The functional test shall verify that no fuses are blown for unprogrammed devices or that the altered item drawing pattern exists for programmed devices, $V_{IH}$ = 3.0 V, $V_{IL}$ = GND.

3/ The test shall check all inputs, gates, and outputs that have programmed. The test shall be performed $V_{CC}$ = 4.5 V and 5.5 V.

4/ For programmed devices only, the outputs are loaded in accordance with figure 4.

5/ Asynchronous output with internal output enable.

6/ For programmed devices only, external output enable set ($\overline{OE}$ = GND).

7/ For programmed devices only, external output enable set ($\overline{OE}$ = 2.0 V).

8/ Synchronous registered output with external output enable (pin 11).

9/ Maximum clock frequency ($f_{MAX}$) is considered met if the proper output state changes occur with the frequency set to the minimum limit given to table III, or $f_{MAX} = 1/(t_{SU} + t_{P(CL)}$.

10/ $t_{SU}$ and $t_{p(CL)}$ may be tested concurrently with $t_{PCL}$ and $t_{PCH}$ by using values within the maximum test values of table III.

11/ For circuit C $I_{OS}$ test may be replaced with an equivalent $I_O$ test as follows: pins 12-19, delete GND and substitute 2.25 V: pin 20 $V_{CC}$, delete 5.0 V, substitute 5.5 V; and in the limits columns, delete min/max of -30/-250 and substitute -15/-125.



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4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535 and as follows. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.6 Programming / verifying procedure for circuit C. The programming specifications on figures 5 and 6, and table IV for circuit C, and the following procedures shall be used for programming the device.

4.6.1 Preverification.

a. Raise $V_{CC}$ to 5.0 V.

b. Raise output disable pin, OD, to $V_{IHH}$.

c. Select an input line by specifying inputs and L/R as shown in table V for circuit C.

d. Select a product line by specifying $A_0$, $A_1$, and $A_2$ one-of-eight select as shown in table VI for circuit C.

e. Pulse the clock pin and verify (with clock at $V_{IL}$) that the output pins, $O_0$ through $O_3$, are in the state corresponding to an unblown fuse. Use the minimum timing conditions as specified on figure 6 for circuit C.
   (1) For verified unblown conditions, continue procedure from c through e.
   (2) For verified blown conditions, stop procedure.

4.6.2 Programming algorithm.

a. Raise output disable pin, OD, to $V_{IHH}$.

b. Lower clock pin to $V_{IL}$.

c. Select an input line as specified in 4.6.1c.

d. Select a product line as specified in 4.6.1d.

e. Raise $V_{CC}$ to $V_{IHH}$.

f. Program the fuse by pulsing the output pins of the selected group – one at a time – to $V_{IHH}$, see figure 6 for circuit C.

g. Lower $V_{CC}$ to 5.0 V.

h. Repeat this procedure from c until pattern is complete.



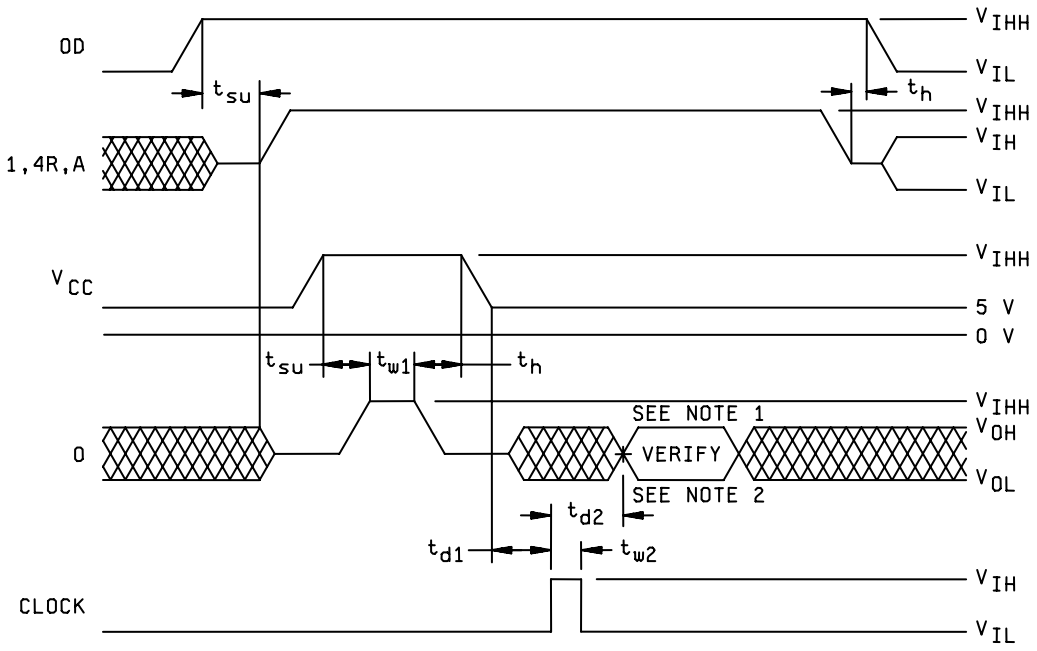
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| Pin No. | Product terms 0 thru 31 | Product terms 32 thru 63 |
|---|---|---|
| 1 | OD | CLOCK |
| 2 | $I_0$ | $I_0$ |
| 3 | $I_1$ | $I_1$ |
| 4 | $I_2$ | $I_2$ |
| 5 | $I_3$ | $I_3$ |
| 6 | $I_4$ | $I_4$ |
| 7 | $I_5$ | $I_5$ |
| 8 | $I_6$ | $I_6$ |
| 9 | $I_7$ | $I_7$ |
| 10 | GND | GND |
| 11 | CLOCK | OD |
| 12 | L/R | $O_3$ |
| 13 | $A_2$ | $O_2$ |
| 14 | $A_1$ | $O_1$ |
| 15 | $A_0$ | $O_0$ |
| 16 | $O_3$ | $A_2$ |
| 17 | $O_2$ | $A_1$ |
| 18 | $O_1$ | $A_0$ |
| 19 | $O_0$ | L/R |
| 20 | $V_{CC}$ | $V_{CC}$ |

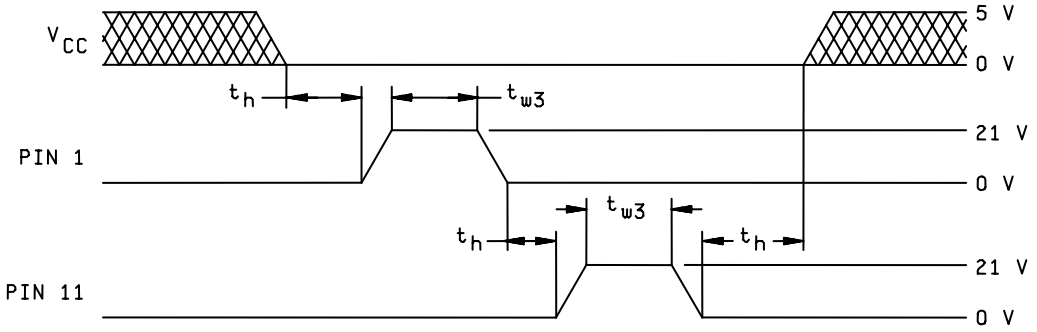
FIGURE 5. Programming pinouts



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PROGRAMMING WAVEFORMS




SECURITY FUSE PROGRAMMING

NOTES:
1. A high level during verify interval indicates that programming has not been successful.
2. A low level during verify interval indicates that programming has been successful.

FIGURE 7. Programming waveforms.



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TABLE IV. Programming characteristics.

| Symbol | Parameters | | Limits | | | Unit |
|---|---|---|---|---|---|---|
| | | | Min | Typ | Max | |
| $V_{CC}$ | Verify-level supply voltage | | 4.5 | 5.0 | 5.5 | V |
| $V_{IH}$ | High-level input voltage | | 2.0 | | 5.5 | V |
| $V_{IL}$ | Low-level input voltage | | | | 0.8 | V |
| $V_{IHH}$ | Program-pulse input voltage | | 10.25 | 10.5 | 10.75 | V |
| $I_{IHH}$ | Program-pulse input current | $O_0$ to $O_3$ | | 20 | 50 | mA |
| | | OD, L/R | | 10 | 25 | |
| | | All other inputs | | 1.5 | 5 | |
| | | $V_{CC}$ | | 250 | 500 | |
| $t_{W1}$ | Program-pulse duration at $O_0$ to $O_3$ pins | | 10 | | 50 | µs |
| $t_{W2}$ | Pulse duration at clock | | 100 | | | ns |
| | Program-pulse duty cycle at $O_0$ to $O_3$ pins | | | | 25 | % |
| $t_{SU}$ | Setup time | | 100 | | | ns |
| $t_h$ | Hold time | | 100 | | | ns |
| $t_{d1}$ | Delay time from $V_{CC}$ to 5 V to clock ↑ | | 100 | | | µs |
| $t_{d2}$ | Delay time from clock ↑ to valid output | | 200 | | | ns |
| | Input voltage at pins 1 and 11 to open verify-protect (security) fuse | | 20 | 21 | 22 | V |
| | Input current to open verify-protect (security) fuse | | | | 400 | mA |
| $t_{w3}$ | Pulse duration to open verify-protect (security) fuse | | 20 | | 50 | µs |
| | $V_{CC}$ value during security fuse programming | | | 0 | 0.4 | V |



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TABLE V. Input line select.

| Input line number | Pin name | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | $I_7$ | $I_6$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | L/R |
| 0 | HH | HH | HH | HH | HH | HH | HH | L | Z |
| 1 | HH | HH | HH | HH | HH | HH | HH | H | Z |
| 2 | HH | HH | HH | HH | HH | HH | HH | L | HH |
| 3 | HH | HH | HH | HH | HH | HH | HH | H | HH |
| 4 | HH | HH | HH | HH | HH | HH | L | HH | Z |
| 5 | HH | HH | HH | HH | HH | HH | H | HH | Z |
| 6 | HH | HH | HH | HH | HH | HH | L | HH | HH |
| 7 | HH | HH | HH | HH | HH | HH | H | HH | HH |
| 8 | HH | HH | HH | HH | HH | L | HH | HH | Z |
| 9 | HH | HH | HH | HH | HH | H | HH | HH | Z |
| 10 | HH | HH | HH | HH | HH | L | HH | HH | HH |
| 11 | HH | HH | HH | HH | HH | H | HH | HH | HH |
| 12 | HH | HH | HH | HH | L | HH | HH | HH | Z |
| 13 | HH | HH | HH | HH | H | HH | HH | HH | Z |
| 14 | HH | HH | HH | HH | L | HH | HH | HH | HH |
| 15 | HH | HH | HH | HH | H | HH | HH | HH | HH |
| 16 | HH | HH | HH | L | HH | HH | HH | HH | Z |
| 17 | HH | HH | HH | H | HH | HH | HH | HH | Z |
| 18 | HH | HH | HH | L | HH | HH | HH | HH | HH |
| 19 | HH | HH | HH | H | HH | HH | HH | HH | HH |
| 20 | HH | HH | L | HH | HH | HH | HH | HH | Z |
| 21 | HH | HH | H | HH | HH | HH | HH | HH | Z |
| 22 | HH | HH | L | HH | HH | HH | HH | HH | HH |
| 23 | HH | HH | H | HH | HH | HH | HH | HH | HH |
| 24 | HH | L | HH | HH | HH | HH | HH | HH | Z |
| 25 | HH | H | HH | HH | HH | HH | HH | HH | Z |
| 26 | HH | L | HH | HH | HH | HH | HH | HH | HH |
| 27 | HH | H | HH | HH | HH | HH | HH | HH | HH |
| 28 | L | HH | HH | HH | HH | HH | HH | HH | Z |
| 29 | H | HH | HH | HH | HH | HH | HH | HH | Z |
| 30 | L | HH | HH | HH | HH | HH | HH | HH | HH |
| 31 | H | HH | HH | HH | HH | HH | HH | HH | HH |

L = $V_{IL,}$ H = $V_{IH,}$ HH = $V_{IHH,}$ Z = high impedance (e.g 10 kΩ to 5 V).



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TABLE VI. Product line select.

| Product line number | Pin identification | | | | | | |
|---|---|---|---|---|---|---|---|
| | $O_3$ | $O_2$ | $O_1$ | $O_0$ | $A_2$ | $A_1$ | $A_0$ |
| 0, 32 | Z | Z | Z | HH | Z | Z | Z |
| 1, 33 | Z | Z | Z | HH | Z | Z | HH |
| 2, 34 | Z | Z | Z | HH | Z | HH | Z |
| 3, 35 | Z | Z | Z | HH | Z | HH | HH |
| 4, 36 | Z | Z | Z | HH | HH | Z | Z |
| 5, 37 | Z | Z | Z | HH | HH | Z | HH |
| 6, 38 | Z | Z | Z | HH | HH | HH | Z |
| 7, 39 | Z | Z | Z | HH | HH | HH | HH |
| 8, 40 | Z | Z | HH | Z | Z | Z | Z |
| 9, 41 | Z | Z | HH | Z | Z | Z | HH |
| 10, 42 | Z | Z | HH | Z | Z | HH | Z |
| 11, 43 | Z | Z | HH | Z | Z | HH | HH |
| 12, 44 | Z | Z | HH | Z | HH | Z | Z |
| 13, 45 | Z | Z | HH | Z | HH | Z | HH |
| 14, 46 | Z | Z | HH | Z | HH | HH | Z |
| 15, 47 | Z | Z | HH | Z | HH | HH | HH |
| 16, 48 | Z | HH | Z | Z | Z | Z | Z |
| 17, 49 | Z | HH | Z | Z | Z | Z | HH |
| 18, 50 | Z | HH | Z | Z | Z | HH | Z |
| 19, 51 | Z | HH | Z | Z | Z | HH | HH |
| 20, 52 | Z | HH | Z | Z | HH | Z | Z |
| 21, 53 | Z | HH | Z | Z | HH | Z | HH |
| 22, 54 | Z | HH | Z | Z | HH | HH | Z |
| 23, 55 | Z | HH | Z | Z | HH | HH | HH |
| 24, 56 | HH | Z | Z | Z | Z | Z | Z |
| 25, 57 | HH | Z | Z | Z | Z | Z | HH |
| 26, 58 | HH | Z | Z | Z | Z | HH | Z |
| 27, 59 | HH | Z | Z | Z | Z | HH | HH |
| 28, 60 | HH | Z | Z | Z | HH | Z | Z |
| 29, 61 | HH | Z | Z | Z | HH | Z | HH |
| 30, 62 | HH | Z | Z | Z | HH | HH | Z |
| 31, 63 | HH | Z | Z | Z | HH | HH | HH |



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4.6.3 First verification pass/for all fuse location.

a. Raise $V_{CC}$ to 4.5 V.

b. Select input line as specified in 4.6.1c.

c. Select a product line as specified in 4.6.1d.

d. Pulse the clock pin and verify (with clock at $V_{IL}$) that the output pins, $O_0$ through $O_3$ are in the correct state.

(1) For verified output state, continue procedure.

(2) For overblown conditions, stop procedure.

(3) For underblown condition, reject part.

e. Repeat this procedure from b until the entire array is exercised.

4.6.4 High voltage verify/for all fuse location.

a. Raise $V_{CC}$ to 5.5 V.

b. Repeat from 4.6.3b to 4.6.3d until the entire array is exercised.

Programming should be attempted at room temperatures, 15°C to 30°C. To prevent further programming, security fuses may be blown. Raise $V_{CC}$ to 6.0 V, and raise pins 1 and 11 to $V_P$ for 10 ms to 1 s.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.



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f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirement for programming the device, including processing option.

j. Requirements for "JAN" marking.

k. Packaging Requirements (see 5.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43218-3990.

6.4 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

GND ..... Electrical ground (common terminal).
$I_{IN}$ ..... Current flowing into an input terminal.
L/R ..... Programming left/right identifier.
$V_{IC}$ ..... Input clamp voltage
$V_{IN}$ ..... Voltage level at an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.4). Longer length leads and lead forming should not affect the part number. It is intended that spare devices for logistic support be acquired in the unprogrammed condition (see 3.8.1) and programmed by the maintenance activity, except where use of quantities for devices with a specific program or pattern justify stocking of preprogrammed devices.



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6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

| Military device type | Generic-industry Type | Circuit designator | Fusible Link | Symbol CAGE numbe |
|---|---|---|---|---|
| 01 | PAL16L8-20<br>Texas Instruments | C | TiW | CGO/01295 |
| 02 | PAL16R8-20<br>Texas Instruments | C | TiW | --- |
| 03 | PAL16R6-20<br>Texas Instruments5 | C | TiW | --- |
| 04 | PAL16R4-20<br>Texas Instruments | C | TiW | --- |
| 05 | PAL16L8-30<br>Texas Instruments | C | TiW | --- |
| 06 | PAL16R8-30<br>Texas Instruments | C | TiW | --- |
| 07 | PAL16R6-30<br>Texas Instruments | C | TiW | --- |
| 08 | PAL16R4-30<br>Texas Instruments | C | TiW | --- |
| 09 | PAL16L8-15<br>Texas Instruments | C | TiW | --- |
| 10 | PAL16R8-15<br>Texas Instruments | C | TiW | --- |
| 11 | PAL16R6-15<br>Texas Instruments | C | TiW | --- |
| 12 | PAL16R4-15<br>Texas Instruments | C | TiW | --- |

6.8 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – SM, MI
Navy - AS, CG, MC, SH TD
Air Force – 03, 19, 99

(Project 5962-2005-043)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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